

48. SD/MMC Host Interface (SDHI)

48.1 Overview

The Secure Digital Host Interface (SDHI) and MultiMediaCard (MMC) Interface provide the functionality required to connect a variety of external memory cards with the MCU. The SDHI supports both 1- and 4-bit buses for connecting different memory cards that support SD, SDHC, and SDXC formats. When developing host devices that are compliant with the SD Specifications, you must comply with the SD Host/Ancillary Product License Agreement (SD HALA).

The MMC interface supports 1-bit, 4-bit, and 8-bit MMC buses that provide eMMC 4.51 (JEDEC Standard JESD 84- B451) device access. This interface also provides backward compatibility and supports for high-speed SDR transfer modes.

Table 48.1 lists the SD/MMC Host Interface specifications and Figure 48.1 shows a block diagram.

Table 48.1 SD/MMC Host Interface specifications

Interface	Parameter	Specifications
SD	SD bus interface	- Compatible with SD memory card and SDIO card - Transfer bus mode selectable from 4-bit wide bus mode or 1-bit default bus mode - Compatible with SD, SDHC, and SDXC formats
	Transfer modes	Default Speed mode, High Speed mode, SDR12 or SDR25
SD and MMC shared	SDHI clock frequency	The SDHI clock is generated by dividing PCLKB by 2^n ($n = 0$ to 9)
	Error check functions	CRC7 (command/response), CRC16 (transfer data)
	Interrupt sources	Card access interrupt (SDHI_MMChn_ACCS), SDIO access interrupt (SDHI_MMChn_SDIO), Card detection interrupt (SDHI_MMChn_CARD) ($n = 0, 1$)
	DMA transfer sources	DMAC and DTC triggerable by the DMA transfer requests (SDHI_MMChn_ODMSDBREQ ($n = 0, 1$)) interrupt SD buffer is read and write accessible using the DMAC
	Other functions	- Card detect function - Write protect support
MMC	MMC bus interface	Transfer bus mode selectable from 1-bit, 4-bit, or 8-bit
	Transfer modes	Backward compatible mode or high-speed SDR mode selectable
	Other functions	e.MMC device access supported
Module-stop function	Module-stop state can be set for each channel to reduce power consumption.	
TrustZone Filter	Security and Privilege attribution can be set for each channel.	

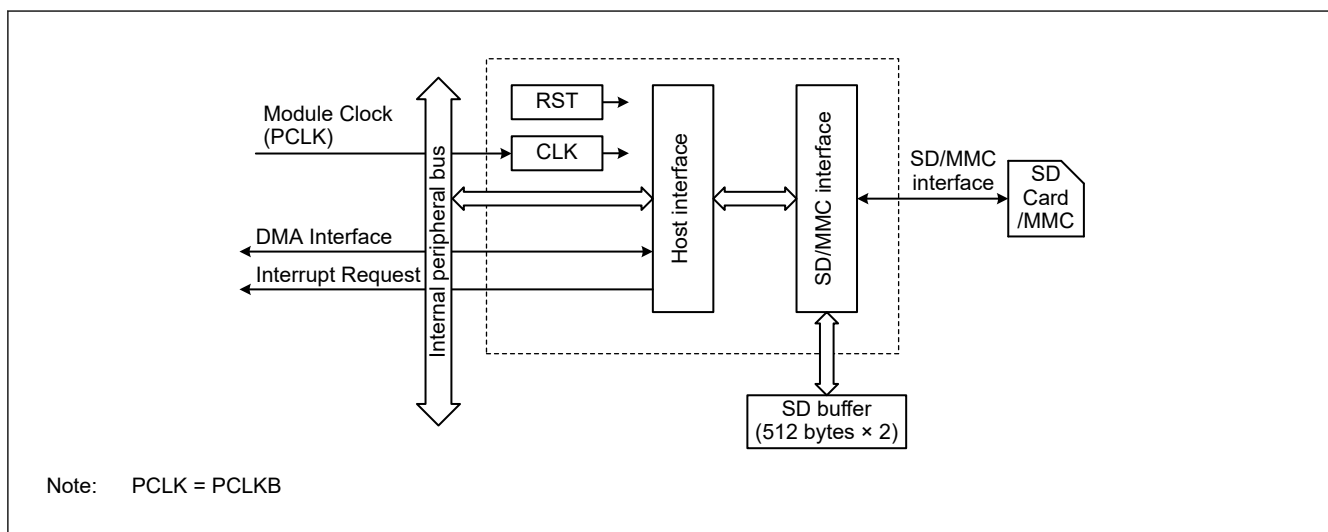


Figure 48.1 SD/MMC Host Interface block diagram

Table 48.2 SDHI I/O pins (n = 0, 1)

Channel	Pin name	I/O	Description
Ch n	SDnCLK	Output	SDHI clock
	SDnCMD	I/O	Command output, response input
	SDnDAT0	I/O	Data 0 (DAT0)
	SDnDAT1	I/O	Data 1 (DAT1), SDIO interrupt
	SDnDAT2	I/O	Data 2 (DAT2), SDIO Read wait
	SDnDAT3	I/O	Data 3 (DAT3), SD Card detect
	SDnDAT4	I/O	MMC Data 4 (DAT4)
	SDnDAT5	I/O	MMC Data 5 (DAT5)
	SDnDAT6	I/O	MMC Data 6 (DAT6)
	SDnDAT7	I/O	MMC Data 7 (DAT7)
	SDnCD	Input	SD card detection
	SDnWP	Input	SD card write protection

48.2 Register Descriptions

48.2.1 SD_CMD : Command Type Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
 SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CMD12AT[1:0]		TRST P	CMDR W	CMDT P	RSPTP[2:0]			ACMD[1:0]		CMDIDX[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	CMDIDX[5:0]	Command Index Field Value Select These bits configure the command index field value. The examples shown include the bit values for the ACMD[1:0] bits. 0x06: CMD6 0x12: CMD18 0x4D: ACMD13	R/W
7:6	ACMD[1:0]	Command Type Select 0 0: CMD 0 1: ACMD Others: Setting prohibited	R/W
10:8	RSPTP[2:0]	Response Type Select*1 0 0 0: Normal mode Depending on the command, the response type and transfer method are selected in the ACMD[1:0] and CMDIDX[5:0] bits. At this time, the values for bits 15 to 11 in this register are invalid. 0 1 1: Extended mode and no response 1 0 0: Extended mode and R1, R5, R6, or R7 response 1 0 1: Extended mode and R1b response 1 1 0: Extended mode and R2 response 1 1 1: Extended mode and R3 or R4 response Others: Setting prohibited	R/W

Bit	Symbol	Function	R/W
11	CMDTP	Data Transfer Select ^{*2} 0: Do not include data transfer (bc, bcr, or ac) in command 1: Include data transfer (adtc) in command	R/W
12	CMDRW	Data Transfer Direction Select ^{*3} 0: Write (SD/MMC Host Interface → SD card/MMC) 1: Read (SD/MMC Host Interface ← SD card/MMC)	R/W
13	TRSTP	Block Transfer Select ^{*3} 0: Single block transfer 1: Multiple blocks transfer	R/W
15:14	CMD12AT[1:0]	CMD12 Automatic Issue Select ^{*4} 0 0: Automatically issue CMD12 during multiblock transfer 0 1: Do not automatically issue CMD12 during multiblock transfer Others: Setting prohibited	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Some commands cannot be used in normal mode.

Note 2. The CMDTP bit is only valid when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b.

Note 3. Bits CMDRW and TRSTP are only valid when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b, and the CMDTP bit is 1.

Note 4. The CMD12AT[1:0] bits are only valid when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b, and the TRSTP bit is 1.

The command type and response type are set in the SD_CMD register. The command type and transfer mode must be set when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b. The sequence starts when a value is written to this register. See [Table 48.8](#) and [Table 48.9](#) for setting examples. Do not write to the SD_CMD register when the SD_INFO2.CBSY flag is 1.

48.2.2 SD_ARG : SD Command Argument Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	—	These bits specify command format[39:8] (argument).	R/W

Note: S-TYPE-3, P-TYPE-3

The SD_ARG register is used for setting the argument field value. Set the SD_ARG register before setting the SD_CMD register. The argument field value of the automatically issued CMD12 is 0x0000_0000 regardless of the SD_ARG register value.

48.2.3 SD_ARG1 : SD Command Argument Register 1

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x00C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits specify command format[39:24] (argument).	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The SD_ARG1 register is used for setting the argument field value. Set the SD_ARG1 register before setting the SD_CMD register. The argument field value of the automatically issued CMD12 is 0x0000_0000 regardless of the SD_ARG1 register value.

48.2.4 SD_STOP : Data Stop Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SEC	—	—	—	—	—	—	—	STP
Value after reset:	0	0	0	0	0	0	0	0*1	0	0	0	0	0	0	0	0*1

Bit	Symbol	Function	R/W
0	STP	Transfer Stop Data transfer stops when this bit is set to 1.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
8	SEC	Block Count Register Value Select*2 0: Disable SD_SECCNT register value 1: Enable SD_SECCNT register value	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The value is initialized by a reset and also on reset triggered by the SOFT_RST.SDRST flag.

Note 2. Do not rewrite this bit when the SD_INFO2.CBSY flag is 1.

The SD_STOP register stops data transfer. During a multiblock transfer sequence, the SD_SECCNT register value (number of blocks to be transferred) can be set to valid or invalid by setting the SD_STOP register.

STP bit (Transfer Stop)

When the STP bit is set to 1 during multiple block transfer, CMD12 is issued to halt the transfer through the SDHI. However, if a command sequence is halted because of a communications error or timeout, CMD12 is not issued. Although continued buffer access is possible even after STP is set to 1, the buffer access error bit (ILR or ILW) in SD_INFO2 is set accordingly.

When STP is set to 1 during transfer for single block write, the access end flag sets when SD_BUF becomes empty, and CMD12 is not issued. If SD_BUF does contain data, the access end flag sets on completion of reception of the busy state without CMD12 being issued.

When STP is set to 1 during transfer for single block read, the access end flag sets immediately after the STP bit is set, and CMD12 is not issued.

When STP is set to 1 during reception of the busy state after an R1b response, the access end flag sets on completion of reception of the busy state without CMD12 being issued.

When STP is set to 1 after a command sequence is completed, CMD12 is not issued and the access end flag does not set.

Set STP to 1 after the response end flag sets.

Set STP to 0 after the access end flag sets.

SEC bit (Block Count Register Value Select)

When SD_CMD is set in the following section to start the command sequence while the SEC bit is set to 1, CMD12 is automatically issued to stop multiblock transfer with the number of blocks set in SD_SECCNT.

CMD18 or CMD25 in normal mode (SD_CMD[10:8] = 000b)

SD_CMD[15:13] = 001b in extended mode (CMD12 is automatically issued, multiple block transfer)

When the command sequence is halted because of a communications error or timeout, CMD12 is not automatically issued.

48.2.5 SD_SECCNT : Block Count Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	—	These bits set the number of blocks to be transferred.	R/W

Note: S-TYPE-3, P-TYPE-3

When performing a multiblock transfer, SD_SECCNT is a read/write register used to set the number of blocks to be transferred. For example, when the register value is 0x0000_0001, 1 block is transferred. When the register value is 0x0000_FFFF, 65,535 blocks are transferred and when the register value is 0xFFFF_FFFF, 4,294,967,295 blocks are transferred. Do not set this register to 0x0000_0000. Do not rewrite the SD_SECCNT register when the SD_INFO2.CBSY flag is 1.

48.2.6 SD_RSPi : SD Card Response Register i (i = 10, 32, 54)

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x018 (SD_RSP10)
0x020 (SD_RSP32)
0x028 (SD_RSP54)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	—	These bits store the response from the SD card/MMC.	R

Note: S-TYPE-3, P-TYPE-3

48.2.7 SD_RSPj : SD Card Response Register j (j = 1, 3, 5)

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x01C (SD_RSP1)
0x024 (SD_RSP3)
0x02C (SD_RSP5)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits store the response from the SD card/MMC.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.8 SD_RSP76 : SD Card Response Register 76

Base address: SDHIn = $0x4025_2000 + 0x0400 \times n$ ($n = 0, 1$)
 SDHIn_NS = $0x5025_2000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x030

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	SD_RSP76[23:16]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SD_RSP76[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
23:0	SD_RSP76[23:0]	These bits store the response from the SD card/MMC.	R
31:24	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.9 SD_RSP7 : SD Card Response Register 7

Base address: SDHIn = $0x4025_2000 + 0x0400 \times n$ ($n = 0, 1$)
 SDHIn_NS = $0x5025_2000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x034

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	SD_RSP7[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	SD_RSP7[7:0]	These bits store the response from the SD card/MMC.	R
31:8	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

SD_RSP10, SD_RSP32, SD_RSP54, SD_RSP1, SD_RSP3, SD_RSP5, SD_RSP76, and SD_RSP7 are read-only registers that store the response from the SD card/MMC. Depending on the type of response from the SD card/MMC, the SD/MMC Host Interface divides and stores the response among the four registers.

Table 48.3 lists the correspondence between the response type and its storage destination.

Table 48.3 Correspondence between response type and storage destination (1 of 2)

Response type	SD_RSP10 register	SD_RSP32 register	SD_RSP54 register	SD_RSP1 register	SD_RSP3 register	SD_RSP5 register	SD_RSP76 register	SD_RSP7 register
R1	[39:8]	—	[39:8]*1	—	—	—	—	—
R1b	[39:8]	—	[39:8]*1	—	—	—	—	—
R2	[39:8]	[71:40]	[103:72]	—	—	—	[127:104]	—
R3	[39:8]	—	—	—	—	—	—	—
R4	[39:8]	—	—	—	—	—	—	—

Table 48.3 Correspondence between response type and storage destination (2 of 2)

Response type	SD_RSP10 register	SD_RSP32 register	SD_RSP54 register	SD_RSP1 register	SD_RSP3 register	SD_RSP5 register	SD_RSP76 register	SD_RSP7 register
R5	[39:8]	—	—	—	—	—	—	—
R6	[39:8]	—	—	—	—	—	—	—
R7	[39:8]	—	—	—	—	—	—	—

Note 1. The responses for CMD18 and CMD25 are stored in registers SD_RSP10 and SD_RSP54. Therefore, even if the SD_RSP10 register is overwritten with the response for the automatically issued CMD12, the response for CMD18 or CMD25 can be confirmed by reading the SD_RSP54 register.

48.2.10 SD_INFO1 : SD Card Interrupt Flag Register 1

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x038

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	SDD3 MON	SDD3I N	SDD3 RM	SDWP MON	—	SDCD MON	SDCDI N	SDCD RM	ACEN D	—	RSPE ND
Value after reset:	0	0	0	0	0	x	0	0	x	0	x	0	0	0 ^{*1}	0	0 ^{*1}

Bit	Symbol	Function	R/W
0	RSPEND	Response End Detection Flag 0: Response end not detected 1: Response end detected	R/(W) ^{*2}
1	—	This bit is read as 0. The write value should be 0.	R/W
2	ACEND	Access End Detection Flag 0: Access end not detected 1: Access end detected	R/(W) ^{*2}
3	SDCDRM	SDnCD Removal Flag 0: SD card/MMC removal not detected by the SDnCD pin 1: SD card/MMC removal detected by the SDnCD pin	R/(W) ^{*2}
4	SDCDIN	SDnCD Insertion Flag 0: SD card/MMC insertion not detected by the SDnCD pin 1: SD card/MMC insertion detected by the SDnCD pin	R/(W) ^{*2}
5	SDCDMON	SDnCD Pin Monitor Flag 0: SDnCD pin level is high ^{*3} 1: SDnCD pin level is low ^{*3}	R
6	—	This bit is read as 0. The write value should be 0.	R/W
7	SDWPMON	SDnWP Pin Monitor Flag 0: SDnWP pin level is high 1: SDnWP pin level is low	R
8	SDD3RM	SDnDAT3 Removal Flag 0: SD card/MMC removal not detected by the SDnDAT3 pin 1: SD card/MMC removal detected by the SDnDAT3 pin	R/(W) ^{*2}
9	SDD3IN	SDnDAT3 Insertion Flag 0: SD card/MMC insertion not detected by the SDnDAT3 pin 1: SD card/MMC insertion detected by the SDnDAT3 pin	R/(W) ^{*2}
10	SDD3MON	SDnDAT3 Pin Monitor Flag 0: SDnDAT3 pin level is low 1: SDnDAT3 pin level is high	R

Bit	Symbol	Function	R/W
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The value is initialized by a reset and also on reset triggered by the `SOFT_RST.SDRST` flag.

Note 2. The flag does not change even if set to 1. Writing 0 changes the flag value to 0.

Note 3. The flag changes when the pin level continues for the period set in the `SD_OPTION.CTOP[3:0]` bits or longer.

The `SD_INFO1` register indicates the detection of a response end or access end for a command sequence. The `SD_INFO1` register also indicates the detection SD card/MMC insertion/removal and the write protection status.

During a multiblock transfer sequence, if `CMD12` or `CMD52` (SDIO abort) is issued, the `ACEND` flag sets to 1, but the `RSPEND` flag remains set to 0.

If the command sequence is stopped because of a communication error or timeout, the `ACEND` flag or `RSPEND` flag sets to 1.

After a reset is canceled, the `SDD3MON` bit, `SDD3IN` flag, and `SDD3RM` flag values are changed in accordance with the status of the `SDnDAT3` ($n = 0, 1$) pin, and their values are changed when data is being transferred in wide bus mode. These 3 bits are used only for SD card. Set flags to be cleared to 0. Set flags that are not being cleared to 1.

RSPEND flag (Response End Detection Flag)

The `RSPEND` flag indicates that a response end was detected.

[Setting conditions]

- When reception of the response is completed
- When transmission of a command without response is completed
- When reception of the busy state after `R1b` response is completed
- When reception of the response to `CMD52` that was issued by setting the `C52PUB` bit to 1 is completed for transfer of multiple block read
- When reception of the response to `CMD52` that was issued by setting the `C52PUB` bit to 1 is completed for transfer of multiple block write
- This bit is set when a command sequence is halted because of a communications error or timeout.

[Clearing conditions]

- When 0 is written to `RSPEND`
- When a command without data is issued.

Note: When a command is issued in absence of data transfer, the `RSPEND` flag becomes 1 after the command sequence ends.

ACEND flag (Access End Detection Flag)

The `ACEND` flag indicates that an access end was detected.

[Setting conditions]

- When read access to the buffer is completed for transfer of single block read
- When read access to the buffer for the last block of data is completed for transfer of multiple block read
- When read access to the buffer and reception of the response to `CMD12` are completed for transfer of multiple block read with automatic issuing of `CMD12`
- When reception of the busy state after reception of the CRC status is completed for transfer of single block write
- When reception of the busy state after reception of the CRC status of the last block of data is completed for transfer of multiple block write
- When reception of the response busy state for `CMD12` is completed for transfer of multiple block write with automatic issuing of `CMD12`
- When reception of the response to `CMD12` that was issued by setting the `STP` bit to 1 is completed for transfer of multiple block read

- When reception of the response busy state for CMD12 that was issued by setting the STP bit to 1 is completed for transfer of multiple block write
- When reception of the response to CMD52 that was issued by setting the IOABT bit to 1 is completed for transfer of multiple block read
- When reception of the response to CMD52 that was issued by setting the IOABT bit to 1 is completed for transfer of multiple block write
- This bit is set when a command sequence is halted because of a communications error or timeout.

[Clearing conditions]

- When 0 is written to ACEND
- When the access end bit is set to 1.

Note: The ACEND flag becomes 1 after the command sequence ends.

SDCDRM flag (SDnCD Removal Flag)

The SDCDRM flag indicates that SDnCD was removed.

[Setting condition]

- After a change in SDnCD from 0 to 1, Mcycle elapsed with SDnCD held at 1.

[Clearing conditions]

- When 0 is written to SDCDRM.

Note: Mcycle is set in bits [3:0] in SD_OPTION.

SDCDIN flag (SDnCD Insertion Flag)

The SD CDIN flag indicates that SDnCD was inserted.

[Setting condition]

- After a change in SDnCD from 1 to 0, Mcycle elapsed with SDnCD held at 0.

[Clearing conditions]

- When 0 is written to SD CDIN.

Note: Mcycle is set in bits [3:0] in SD_OPTION.

SDD3RM flag (SDnDAT3 Removal Flag)

The SDD3RM flag indicates that SDnDAT3 was removed.

[Setting condition]

- After a change in SDnDAT3 from 1 to 0, two cycles of PCLKB elapsed with SDnDAT3 held at 0.

[Clearing condition]

- When 0 is written to SDD3RM.

SDD3IN flag (SDnDAT3 Insertion Flag)

The SDD3IN flag indicates that SDnDAT3 was inserted.

[Setting condition]

- After a change in SDnDAT3 from 0 to 1, two cycles of PCLKB elapsed with SDnDAT3 held at 1.

[Clearing condition]

- When 0 is written to SDD3IN.

48.2.11 SD_INFO2 : SD Card Interrupt Flag Register 2

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x03C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ILA	CBSY	SD_CLK_CTRLLEN	—	—	—	BWE	BRE	SDD0MON	RSPTO	ILR	ILW	DTO	ENDE	CRCE	CMDE
Value after reset:	0 ^{*1}	0 ^{*1}	1 ^{*1}	0	x	0	0 ^{*1}	0 ^{*1}	x	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}

Bit	Symbol	Function	R/W
0	CMDE	Command Error Detection Flag 0: Command error not detected 1: Command error detected	R/W ^{*2}
1	CRCE	CRC Error Detection Flag 0: CRC error not detected 1: CRC error detected	R/W ^{*2}
2	ENDE	End Bit Error Detection Flag 0: End bit error not detected 1: End bit error detected	R/W ^{*2}
3	DTO	Data Timeout Detection Flag 0: Data timeout not detected 1: Data timeout detected	R/W ^{*2}
4	ILW	SD_BUF0 Illegal Write Access Detection Flag 0: Illegal write access to the SD_BUF0 register not detected 1: Illegal write access to the SD_BUF0 register detected	R/W ^{*2}
5	ILR	SD_BUF0 Illegal Read Access Detection Flag 0: Illegal read access to the SD_BUF0 register not detected 1: Illegal read access to the SD_BUF0 register detected	R/W ^{*2}
6	RSPTO	Response Timeout Detection Flag 0: Response timeout not detected 1: Response timeout detected	R/W ^{*2}
7	SDD0MON	SDnDAT0 Pin Status Flag 0: SDnDAT0 pin is low 1: SDnDAT0 pin is high	R
8	BRE	SD_BUF0 Read Enable Flag 0: Disable read access to the SD_BUF0 register 1: Enable read access to the SD_BUF0 register	R/W ^{*2}
9	BWE	SD_BUF0 Write Enable Flag 0: Disable write access to the SD_BUF0 register 1: Enable write access to the SD_BUF0 register	R/W ^{*2}
10	—	This bit is read as 0. The write value should be 0.	R/W
11	—	The read value is undefined. The write value should be 1.	R/W
12	—	This bit is read as 0. The write value should be 0.	R/W
13	SD_CLK_CTRLLEN	SD_CLK_CTRL Write Enable Flag 0: SD/MMC bus (CMD and DAT lines) is busy, so write access to the SD_CLK_CTRL.CLKEN and CLKSEL[7:0] bits is disabled 1: SD/MMC bus (CMD and DAT lines) is not busy, so write access to the SD_CLK_CTRL.CLKEN and CLKSEL[7:0] bits is enabled	R

Bit	Symbol	Function	R/W
14	CBSY	Command Sequence Status Flag 0: Command sequence complete 1: Command sequence in progress (busy)	R
15	ILA	Illegal Access Error Detection Flag 0: Illegal access error not detected 1: Illegal access error detected	R/W ²
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The value is initialized by a reset and also on reset triggered by the `SOFT_RST.SDRST` flag.

Note 2. The flag does not change even if set to 1. Writing 0 changes the flag value to 0.

The `SD_INFO2` register indicates the status of the SD buffer and the status of the SD card/MMC. Set flags to be cleared to 0. Set flags that are not being cleared to 1.

CMDE flag (Command Error Detection Flag)

The CMDE flag indicates that a command error was detected. The command sequence stops when a command error occurs. When the `SDIO_MODE.C52PUB` bit is set to 1 and `CMD52` is automatically issued, if a communication error or response timeout occurs, the command sequence is not completed. Perform the error processing shown in [section 48.3.12. IO_RW_EXTENDED Command \(SD: CMD53/Multiple Block Read\)](#) or [section 48.3.13. IO_RW_EXTENDED Command \(SD: CMD53/Multiple Block Write\)](#), and complete the command sequence.

[Setting conditions]

- The command index of the transmitted command differs from the command index of the received response.
- The command index of a command issued within a command sequence differs from the command index of the received response.

[Clearing condition]

- When 0 is written to CMDE.

CRCE flag (CRC Error Detection Flag)

The CRCE flag indicates that a CRC error was detected. The command sequence stops when a CRC error occurs. When the `SDIO_MODE.C52PUB` bit is set to 1 and `CMD52` is automatically issued, if a communication error or response timeout occurs, the command sequence is not completed. Perform the error processing shown in [section 48.3.12. IO_RW_EXTENDED Command \(SD: CMD53/Multiple Block Read\)](#) or [section 48.3.13. IO_RW_EXTENDED Command \(SD: CMD53/Multiple Block Write\)](#), and complete the command sequence.

[Setting conditions]

- When an error occurs in the CRC status.
- When a CRC error occurs in the read data.
- When a CRC error occurs in the response.
- A CRC error in the response to a command issued within a command sequence.

[Clearing condition]

- When 0 is written to CRCE.

ENDE flag (End Bit Error Detection Flag)

The ENDE flag indicates that an end bit error was detected. The command sequence is stopped when an end bit error occurs. When the `SDIO_MODE.C52PUB` bit is set to 1 and `CMD52` is automatically issued, if a communication error or response timeout occurs, the command sequence is not completed. Perform the error processing shown in [section 48.3.12. IO_RW_EXTENDED Command \(SD: CMD53/Multiple Block Read\)](#) or [section 48.3.13. IO_RW_EXTENDED Command \(SD: CMD53/Multiple Block Write\)](#), and complete the command sequence.

[Setting conditions]

- When an error occurs in the response length (and the end bit is not detected).
- When an error occurs in the read data length (and the end bit is not detected among the valid bits).

- When an error occurs in the CRC status length (and the end bit is not detected).
- An error in the length of a response to a command issued within a command sequence, for example when the end bit is not detected.

[Clearing condition]

- When 0 is written to ENDE.

DTO flag (Data Timeout Detection Flag)

The DTO flag indicates that a data timeout was detected. The command sequence stops when a data timeout occurs.

[Setting conditions]

- After R1b response, the busy state (SDnDAT0 = 0) continues for longer than Ncycle.
- After CRC status, the busy state (SDnDAT0 = 0) continues for longer than Ncycle.
- After write data, the CRC status is not received though Ncycle has elapsed.
- After read command, read data is not received though a time longer than Ncycle has elapsed.
- After CMD12 is issued within a command sequence, the busy state (SDnDAT0 = 0) for longer than Ncycle continues.
- After the reception of read data, read data for the next block are not received though a time longer than Ncycle has elapsed.
- After release of the read wait state, read data for the next block are not received though a time longer than Ncycle has elapsed.

Note: Ncycle is set in bits [7:4] in SD_OPTION.

[Clearing condition]

- When 0 is written to DTO.

ILW flag (SD_BUF0 Illegal Write Access Detection Flag)

The ILW flag indicates that an SD_BUF0 illegal write access was detected.

[Setting conditions]

- When data is written to SD_BUF0 while it is not in the data read/write command state.
- When data is written to SD_BUF0 while SD_BUF is full.
- When data is written to SD_BUF0 while an error occurs in the CRC status or CRC status length.
- When data is written to SD_BUF0 while a busy state after the CRC status continues for longer than Ncycle.

Note: Ncycle is set in bits [7:4] in SD_OPTION.

[Clearing condition]

- When 0 is written to ILW.

ILR flag (SD_BUF0 Illegal Read Access Detection Flag)

The ILR flag indicates that an SD_BUF0 illegal read access was detected.

[Setting conditions]

- When SD_BUF is empty while SD_BUF0 is read.
- When data with a CRC error or END error is read from SD_BUF0.

[Clearing condition]

- When 0 is written to ILR.

RSPTO flag (Response Timeout Detection Flag)

The RSPTO flag indicates that a response timeout was detected. The command sequence is stopped when a response timeout occurs. When the SDIOMD.C52PUB bit is set to 1 and CMD52 is automatically issued, if a communication error

or response timeout occurs, the command sequence is not completed. Perform the error processing shown in [section 48.3.12. IO_RW_EXTENDED Command \(SD: CMD53/Multiple Block Read\)](#) or [section 48.3.13. IO_RW_EXTENDED Command \(SD: CMD53/Multiple Block Write\)](#), and complete the command sequence.

[Setting condition]

- When a response is not received though a time longer than 640 cycles of SD/MMC clock has elapsed (including a response to a command issued within a command sequence).

[Clearing condition]

- When 0 is written to RSPTO.

SDD0MON flag (SDnDAT0 Pin Status Flag)

The SDD0MON flag indicates the status of the SDnDAT0 pin. If the data timeout (DTO) is set but the response timeout (RSPTO) is not set after the Erase command is issued, the end of the Erase sequence (SDD0MON = 1) is confirmed by polling DAT0.

If a communication error or timeout occurs during a write sequence, the DAT0 bit might retain the value 0.

While the SD/MMC clock is stopped, the DAT0 bit retains the value before the clock is stopped.

BRE flag (SD_BUF0 Read Enable Flag)

The BRE flag indicates that SD_BUF0 is enabled for reading.

[Setting conditions]

- When data set in SD_SIZE is stored in SD_BUF0 at single block transfer.
- When data set in SD_SIZE is stored in either bank 1 or bank 2 of SD_BUF0 at multiple block transfer.

[Clearing conditions]

- When 0 is written to BRE
- Reading of a block of data from SD_BUF0 by DMA transfer

When data is read from SD_BUF0 by the CPU, clear BRE then read the amount of data specified in SD_SIZE.

Even if a CRC error or an END error occurs while block data is read, data is stored in SD_BUF0 and BRE is set.

BWE flag (SD_BUF0 Write Enable Flag)

The BWE flag indicates that SD_BUF0 is enabled for writing.

[Setting conditions]

- When SD_BUF0 is empty at single block transfer.
- When either bank 1 or bank 2 of SD_BUF0 is empty at multiple block transfer.

[Clearing conditions]

- When 0 is written to BWE.
- Writing of a block of data to SD_BUF0 by DMA transfer.

When data is written to SD_BUF0 by the CPU, clear BWE and then write the amount of data specified in SD_SIZE.

SD_CLK_CTRLLEN flag (SD_CLK_CTRL Write Enable Flag)

When a command sequence is started by writing to SD_CMD, the CBSY bit is set to 1 and, at the same time, the SD_CLK_CTRLLEN bit is set to 0. The SD_CLK_CTRLLEN bit is set to 1 after 8 cycles of SDCLK have elapsed after the CBSY bit clears to 0 on completion of the command sequence.

ILA flag (Illegal Access Error Detection Flag)

The ILA flag indicates that an illegal access error was detected.

[Setting conditions]

- Writing of data to SD_CMD within a command sequence (CBSY = 1).

- When SD_CMD[11] = 1 (command with data transfer) and SD_CMD[7:0] = 0000 1100b (CMD12) are set in SD_CMD.

[Clearing condition]

- When 0 is written to ILA.

48.2.12 SD_INFO1_MASK : SD INFO1 Interrupt Mask Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x040

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	SDD3I NM	SDD3 RMM	—	—	—	SDCDI NM	SDCD RMM	ACEN DM	—	RSPE NDM
Value after reset:	0	0	0	0	0	0	1	1	0	0	0	1	1	1	0	1

Bit	Symbol	Function	R/W
0	RSPENDM	Response End Interrupt Request Mask 0: Do not mask response end interrupt request 1: Mask response end interrupt request	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
2	ACENDM	Access End Interrupt Request Mask 0: Do not mask access end interrupt request 1: Mask access end interrupt request	R/W
3	SDCDRMM	SDnCD Removal Interrupt Request Mask 0: Do not mask SD card/MMC removal interrupt request by the SDnCD pin 1: Mask SD card/MMC removal interrupt request by the SDnCD pin	R/W
4	SDCDINM	SDnCD Insertion Interrupt Request Mask 0: Do not mask SD card/MMC insertion interrupt request by the SDnCD pin 1: Mask SD card/MMC insertion interrupt request by the SDnCD pin	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	SDD3RMM	SDnDAT3 Removal Interrupt Request Mask 0: Do not mask SD card/MMC removal interrupt request by the SDnDAT3 pin 1: Mask SD card/MMC removal interrupt request by the SDnDAT3 pin	R/W
9	SDD3INM	SDnDAT3 Insertion Interrupt Request Mask 0: Do not mask SD card/MMC insertion interrupt request by the SDnDAT3 pin 1: Mask SD card/MMC insertion interrupt request by the SDnDAT3 pin	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The SD_INFO1_MASK register enables or disables interrupt requests from the status flags in the SD_INFO1 register. See [Table 48.5](#), for details on the relationship between the status flags and the requested interrupt source.

48.2.13 SD_INFO2_MASK : SD INFO2 Interrupt Mask Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
 SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x044

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ILAM	—	—	—	—	—	BWEM	BREM	—	RSPTOM	ILRM	ILWM	DTOM	ENDEM	CRCEM	CMDEM
Value after reset:	1	0	0	0	1	0	1	1	0	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	CMDEM	Command Error Interrupt Request Mask 0: Do not mask command error interrupt request 1: Mask command error interrupt request	R/W
1	CRCEM	CRC Error Interrupt Request Mask 0: Do not mask CRC error interrupt request 1: Mask CRC error interrupt request	R/W
2	ENDEM	End Bit Error Interrupt Request Mask 0: Do not mask end bit detection error interrupt request 1: Mask end bit detection error interrupt request	R/W
3	DTOM	Data Timeout Interrupt Request Mask 0: Do not mask data timeout interrupt request 1: Mask data timeout interrupt request	R/W
4	ILWM	SD_BUF0 Register Illegal Write Interrupt Request Mask 0: Do not mask illegal write detection interrupt request for the SD_BUF0 register 1: Mask illegal write detection interrupt request for the SD_BUF0 register	R/W
5	ILRM	SD_BUF0 Register Illegal Read Interrupt Request Mask 0: Do not mask illegal read detection interrupt request for the SD_BUF0 register 1: Mask illegal read detection interrupt request for the SD_BUF0 register	R/W
6	RSPTOM	Response Timeout Interrupt Request Mask 0: Do not mask response timeout interrupt request 1: Mask response timeout interrupt request	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
8	BREM*1	BRE Interrupt Request Mask 0: Do not mask read enable interrupt request for the SD buffer 1: Mask read enable interrupt request for the SD buffer	R/W
9	BWEM*1	BWE Interrupt Request Mask 0: Do not mask write enable interrupt request for the SD_BUF0 register 1: Mask write enable interrupt request for the SD_BUF0 register	R/W
10	—	This bit is read as 0. The write value should be 0.	R/W
11	—	This bit is read as 1. The write value should be 1.	R/W
14:12	—	These bits are read as 0. The write value should be 0.	R/W
15	ILAM	Illegal Access Error Interrupt Request Mask 0: Do not mask illegal access error interrupt request 1: Mask illegal access error interrupt request	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. When the SD_INFO2_MASK.BWEM bit is 0 or the SD_INFO2_MASK.BREM bit is 0, set the SD_DMAEN.DMAEN bit to 0. When the SD_DMAEN.DMAEN bit is 1, set the SD_INFO2_MASK.BWEM bit to 1 and the SD_INFO2_MASK.BREM bit to 1.

The SD_INFO2_MASK register enables or disables interrupt requests from the status flags in the SD_INFO2 register. See [Table 48.5](#) for details on the relationship between the status flags and the requested interrupt source.

48.2.14 SD_CLK_CTRL : SD Clock Control Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x048

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	CLKC TRLE N	CLKE N	CLKSEL[7:0]							
Value after reset:	0	0	0	0	0	0	0	0 ^{*1}	0	0	1	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CLKSEL[7:0]	SDHI Clock Frequency Select ^{*2} 0xFF: PCLKB 0x00: PCLKB/2 0x01: PCLKB/4 0x02: PCLKB/8 0x04: PCLKB/16 0x08: PCLKB/32 0x10: PCLKB/64 0x20: PCLKB/128 0x40: PCLKB/256 0x80: PCLKB/512 Others: Setting prohibited	R/W
8	CLKEN	SD/MMC Clock Output Control ^{*2} 0: Disable SD/MMC clock output (fix SDnCLK signal low) 1: Enable SD/MMC clock output	R/W
9	CLKCTRLN	SD/MMC Clock Output Automatic Control Select 0: Disable automatic control of SD/MMC clock output 1: Enable automatic control of SD/MMC clock output	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The value is initialized by a reset and also on reset triggered by the SOFT_RST.SDRST flag.

Note 2. Bits CLKSEL[7:0] and CLKEN cannot be write accessed when the SD_INFO2.SD_CLK_CTRLN flag is 0.

The SD_CLK_CTRL register controls the SD/MMC clock frequency settings and output. Set the CLKEN bit to 1 before writing to the SD_CMD register to start a command sequence. Do not write to the SD_CLK_CTRL register when the SD_INFO2.SD_CLK_CTRLN flag is 0.

CLKCTRLN bit (SD/MMC Clock Output Automatic Control Select)

The CLKCTRLN bit enables or disables the automatic control function for SD/MMC clock output, which causes the SD/MMC clock to output only within a command sequence.

The timing with which SD/MMC clock output starts and stops is as follows:

- SD/MMC clock output starts after writing to SD_CMD
- SD/MMC clock output stops when 8 cycles of SD/MMC clock have elapsed after the end of the command sequence.

In addition, SD/MMC clock is fixed to 0 while SCLKEN of SD_CLK_CTRL is 0, regardless of the value of this bit.

48.2.15 SD_SIZE : Transfer Data Length Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x04C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	LEN[9:0]									
Value after reset:	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	LEN[9:0]	Transfer Data Size Setting These bits specify the transfer data size.*1	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Do not rewrite these bits when the SD_INFO2.CBSY flag is 1.

The SD_SIZE register sets the transfer data size.

LEN[9:0] bits (Transfer Data Size Setting)

When using single block transfer, the transfer data size can be set in the LEN[9:0] bits from 1 byte to 512 bytes. When CMD12 is automatically issued during a multiblock transfer sequence (CMD18 and CMD25), the transfer data size can only be set to 512 bytes. When CMD12 is not automatically issued during a multiblock transfer sequence, the transfer data size can be set to 32, 64, 128, 256, or 512 bytes. However, a 32, 64, 128, or 256 bytes multiblock read transfer can only be performed during an SDIO multiblock transfer (CMD53). Do not set these bits to 0 when using a command that includes data transfer.

48.2.16 SD_OPTION : SD Card Access Control Option Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x050

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WIDT H	—	WIDT H8	—	—	—	—	TOUT MASK	TOP[3:0]				CTOP[3:0]			
Value after reset:	0*1	1	0*1	0	0	0	0	0*1	1*1	1*1	1*1	0*1	1*1	1*1	1*1	0*1

Bit	Symbol	Function	R/W															
3:0	CTOP[3:0]	Card Detection Time Counter* ² 0x0: PCLKB × 2 ¹⁰ 0x1: PCLKB × 2 ¹¹ 0x2: PCLKB × 2 ¹² 0x3: PCLKB × 2 ¹³ 0x4: PCLKB × 2 ¹⁴ 0x5: PCLKB × 2 ¹⁵ 0x6: PCLKB × 2 ¹⁶ 0x7: PCLKB × 2 ¹⁷ 0x8: PCLKB × 2 ¹⁸ 0x9: PCLKB × 2 ¹⁹ 0xA: PCLKB × 2 ²⁰ 0xB: PCLKB × 2 ²¹ 0xC: PCLKB × 2 ²² 0xD: PCLKB × 2 ²³ 0xE: PCLKB × 2 ²⁴ 0xF: Setting prohibited	R/W															
7:4	TOP[3:0]	Timeout Counter* ² 0x0: SDHI clock × 2 ¹³ 0x1: SDHI clock × 2 ¹⁴ 0x2: SDHI clock × 2 ¹⁵ 0x3: SDHI clock × 2 ¹⁶ 0x4: SDHI clock × 2 ¹⁷ 0x5: SDHI clock × 2 ¹⁸ 0x6: SDHI clock × 2 ¹⁹ 0x7: SDHI clock × 2 ²⁰ 0x8: SDHI clock × 2 ²¹ 0x9: SDHI clock × 2 ²² 0xA: SDHI clock × 2 ²³ 0xB: SDHI clock × 2 ²⁴ 0xC: SDHI clock × 2 ²⁵ 0xD: SDHI clock × 2 ²⁶ 0xE: SDHI clock × 2 ²⁷ 0xF: Setting prohibited	R/W															
8	TOUTMASK	Timeout Mask 0: Activate timeout 1: Inactivate timeout (do not set RSPTO and DTO bits of SD_INFO2 or CRCBSYTO, CRCTO, RDTO, BSYTO1, BSYTO0, RSPTO1 and RSPTO0 bits of SD_ERR_STS2) When timeout occurs because of an inactivated timeout, execute a software reset to terminate the command sequence.	R/W															
12:9	—	These bits are read as 0. The write value should be 0.	R/W															
13	WIDTH8	Bus Width* ² See bit 15 WIDTH bit.	R/W															
14	—	This bit is read as 1. The write value should be 1.	R/W															
15	WIDTH	Bus Width* ² <table><tr><th>WIDTH</th><th>WIDTH8</th><th>Bus Width</th></tr><tr><td>0</td><td>1</td><td>8-bit width</td></tr><tr><td>0</td><td>0</td><td>4-bit width</td></tr><tr><td>1</td><td>0</td><td>1-bit width</td></tr><tr><td>1</td><td>1</td><td>1-bit width</td></tr></table> For 1-byte write transfers, set 4-bit or 1-bit width. Do not set 8-bit width.	WIDTH	WIDTH8	Bus Width	0	1	8-bit width	0	0	4-bit width	1	0	1-bit width	1	1	1-bit width	R/W
WIDTH	WIDTH8	Bus Width																
0	1	8-bit width																
0	0	4-bit width																
1	0	1-bit width																
1	1	1-bit width																
31:16	—	These bits are read as 0. The write value should be 0.	R/W															

Note: S-TYPE-3, P-TYPE-3

Note 1. The value is initialized by a reset and also on reset triggered by the SOFT_RST.SDRST flag.

Note 2. Do not rewrite these bits when the SD_INFO2.CBSY flag is 1.

The SD bus width and timeout counter are set in the SD_OPTION register.

48.2.17 SD_ERR_STS1 : SD Error Status Register 1

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x058

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	CRCTK[2:0]			CRCT KE	RDCR CE	RSPC RCE1	RSPC RCE0	—	—	CRCL ENE	RDLE NE	RSPL ENE1	RSPL ENE0	CMDE 1	CMDE 0
Value after reset:	0	0 ^{*1}	1 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0	0	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}

Bit	Symbol	Function	R/W
0	CMDE0	Command Error Flag 0 0: No error exists in command index field value of a command ^{*2} response 1: Error exists in command index field value of a command ^{*2} response	R
1	CMDE1	Command Error Flag 1 0: No error exists in command index field value of a command ^{*3} response 1: Error exists in command index field value of a command ^{*3} response (with SD_CMD.CMDIDX[5:0] setting, an error that occurs with CMD12 issue is indicated in the CMDE0 flag)	R
2	RSPLNE0	Response Length Error Flag 0 0: No error exists in command ^{*2} response length 1: Error exists in command ^{*2} response length	R
3	RSPLNE1	Response Length Error Flag 1 0: No error exists in command ^{*3} response length 1: Error exists in command ^{*3} response length (with SD_CMD.CMDIDX[5:0] setting, an error that occurs with CMD12 issue is indicated in the RSPLNE0 flag)	R
4	RDLENE	Read Data Length Error Flag 0: No read data length error occurred 1: Read data length error occurred	R
5	CRCLNE	CRC Status Token Length Error Flag 0: No CRC status token length error occurred 1: CRC status token length error occurred	R
7:6	—	These bits are read as 0.	R
8	RSPCRCE0	Response CRC Error Flag 0 0: No CRC error detected in command ^{*2} response 1: CRC error detected in command ^{*2} response	R
9	RSPCRCE1	Response CRC Error Flag 1 0: No CRC error detected in command ^{*3} response (with SD_CMD.CMDIDX[5:0] setting, an error that occurs with CMD12 issue is indicated in the RSPCRCE0 flag) 1: CRC error detected in command ^{*3} response	R
10	RDCRCE	Read Data CRC Error Flag 0: No CRC error detected in read data 1: CRC error detected in read data	R
11	CRCTKE	CRC Status Token Error Flag 0: No error detected in CRC status token 1: Error detected in CRC status token	R

Bit	Symbol	Function	R/W
14:12	CRCTK[2:0]	CRC Status Token These bits store the CRC status token value (normal value is 010b).	R
31:15	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

Note 1. The value is initialized by a reset and also on reset triggered by the `SOFT_RST.SDRST` flag.

Note 2. CMD other than CMD12 when automatic issuing is enabled for multiple block transfer by the setting in `SD_CMD`, CMD12 when the STP bit in `SD_STOP` is set to 1, or CMD52 when the C52PUB or IOABT bit in `SDIO_MODE` is set to 1.

Note 3. CMD12 when automatic issuing is enabled for multiple block transfer by the setting in `SD_CMD`, CMD12 when the STP bit in `SD_STOP` is set to 1, or CMD52 when the C52PUB or IOABT bit in `SDIO_MODE` is set to 1.

The `SD_ERR_STS1` register indicates the CRC status token, CRC error, end bit error, and command error.

48.2.18 SD_ERR_STS2 : SD Error Status Register 2

Base address: `SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)`
`SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)`

Offset address: 0x05C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	CRCB SYTO	CRCT O	RDTO	BSYT O1	BSYT O0	RSPT O1	RSPT O0
Value after reset:	0	0	0	0	0	0	0	0	0	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}	0 ^{*1}

Bit	Symbol	Function	R/W
0	RSPTO0	Response Timeout Flag 0 0: After command ^{*2} was issued, response was received in less than 640 cycles of the SD/MMC clock 1: After command ^{*2} was issued, response was not received in 640 or more cycles of the SD/MMC clock	R
1	RSPTO1	Response Timeout Flag 1 0: After command ^{*3} was issued, response was received in less than 640 cycles of the SD/MMC clock 1: After command ^{*3} was issued, response was not received after 640 or more cycles of the SD/MMC clock (with <code>SD_CMD.CMDIDX[5:0]</code> setting, an error that occurs with CMD12 issue is indicated in the RSPTO0 flag)	R
2	BSYTO0	Busy Timeout Flag 0 0: After R1b response was received, SD/MMC was released from the busy state during the specified period ^{*4} 1: After R1b response was received, SD/MMC was in the busy state after the specified period ^{*4} elapsed	R
3	BSYTO1	Busy Timeout Flag 1 0: After CMD12 was automatically issued, SD/MMC was released from the busy state during the specified period ^{*4} 1: After CMD12 was automatically issued, SD/MMC was in the busy state after the specified period ^{*4} elapsed (with <code>SD_CMD.CMDIDX[5:0]</code> setting, an error that occurs with CMD12 issue is indicated in the BSYTO0 flag)	R
4	RDTO	Read Data Timeout Flag When a read command is issued, this flag sets to 1 when read data is not received after the specified period ^{*4} elapses. When read data is received, this flag sets to 1 when the next block of read data is not received after the specified period ^{*4} elapses. When the SD/MMC exits the read wait state, this flag sets to 1 when the next block of read data is not received after the specified period ^{*4} elapses.	R

Bit	Symbol	Function	R/W
5	CRCTO	CRC Status Token Timeout Flag 0: After CRC data was written to the SD card/MMC, a CRC status token was received during the specified period*4 1: After CRC data was written to the SD card/MMC, a CRC status token was not received after the specified period*4 elapsed	R
6	CRCBSYTO	CRC Status Token Busy Timeout Flag 0: After a CRC status token was received, the SD/MMC was released from the busy state during the specified period*4 1: After a CRC status token was received, the SD/MMC was in the busy state after the specified period*4 elapsed	R
31:7	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

Note 1. The value is initialized by a reset and also on reset triggered by the SOFT_RST.SDRST flag.

Note 2. CMD other than CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD_CMD, CMD12 when the STP bit in SD_STOP is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO_MODE is set to 1.

Note 3. CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD_CMD, CMD12 when the STP bit in SD_STOP is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO_MODE is set to 1.

Note 4. Set the SD_OPTION.TOP[3:0] bits to select the number of *n* cycles.

The SD_ERR_STS2 register indicates the timeout status.

48.2.19 SD_BUF0 : SD Buffer Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × *n* (*n* = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × *n* (*n* = 0, 1)

Offset address: 0x060

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:																
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
31:0	n/a	When writing to the SD card, the write data is written to this register. When reading from the SD card, the read data is read from this register.	R/W

This register is internally connected to two 512-byte buffers.

If both buffers are not empty when executing multiple block read, the SD card/MMC clock is stopped to suspend receiving data. When one of the buffers is empty, the SD card/MMC clock is supplied to resume receiving data.

48.2.20 SDIO_MODE : SDIO Mode Control Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x068

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	C52P UB	IOABT	—	—	—	—	—	RWRE Q	—	INTEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	INTEN	SDIO Interrupt Acceptance Enable*1 0: Disable SDIO interrupt acceptance 1: Enable SDIO interrupt acceptance	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
2	RWREQ	Read Wait Request 0: Allow SD/MMC to exit read wait state 1: Request for SD/MMC to enter read wait state	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
8	IOABT	SDIO Abort If this bit is set to 1 during multiblock transfer triggered by CMD53, CMD52 is immediately issued, and the command sequence is aborted.	R/W
9	C52PUB	SDIO None Abort If this bit is set to 1 during multiblock transfer triggered by CMD53, CMD52 is issued after the transfer process is complete, and the command sequence is completed.	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Do not rewrite this bit when the SD_INFO2.CBSY flag is 1.

The SDIO_MODE register controls reception of the SDIO interrupt, CMD52 issuance during multiblock transfer, and read wait request. Do not set bits C52PUB and IOABT to 1 at the same time.

RWREQ bit (Read Wait Request)

When RWREQ is set to 1 in the CMD53 (multiple block) read sequence, the block transfer enters the read wait state between blocks.

[Read wait state releasing]

- The read wait state is released, when RWREQ is cleared to 0 in the read wait state.
- When IOABT is set to 1 in the read wait state, RWREQ is automatically cleared to 0 after CMD52 is issued, and then the read wait state is released.
- When C52PUB and RWREQ are set to 1 simultaneously in the CMD53 (multiple block) read sequence, the read wait state is not automatically released. Therefore, after the CMD52 response is received, clear RWREQ. You must set RWREQ and C52PUB simultaneously.

When RWREQ is set to 1 while the last block in the CMD53 (multiple block) read sequence is transferred, the read wait state is not entered and RWREQ is automatically cleared to 0 by setting access end. Set RWREQ to 1 after the response end flag sets.

IOABT bit (SDIO Abort)

When the IOABT bit is set to 1 in a CMD53 (multiple block) sequence, the CMD53 sequence is halted and CMD52 is issued. However, if a command sequence is halted because of a communication error or timeout, CMD52 is not issued.

Although continued buffer access is possible even after IOABT is set to 1, the buffer access error bit (ILR or ILW) in SD_INFO2 is set accordingly. Set SD_ARG before setting IOABT to 1.

When IOABT is set to 1 during transfer for a single block write, the access end flag sets when SD_BUF0 becomes empty, and CMD52 is not issued. If SD_BUF0 contains data, the access end flag sets on completion of reception of the busy state without CMD52 being issued.

When IOABT is set to 1 during transfer for single block read, the access end flag sets immediately after IOABT is set, and CMD52 is not issued.

When IOABT is set to 1 during reception of the busy state after an R1b response, the access end flag sets on completion of reception of the busy state without CMD52 being issued.

When IOABT is set to 1 after a command sequence is completed, CMD52 is not issued and the access end flag does not set.

Set IOABT to 1 after the response end flag sets.

Set IOABT to 0 after the access end flag sets.

C52PUB bit (SDIO None Abort)

When the C52PUB bit is set to 1 in the CMD53 (multiple block) write sequence, CMD52 is automatically issued between blocks if SD_BUF0 becomes empty. C52PUB is automatically cleared to 0 after reception of the response to CMD52 is completed. Additionally, if C52PUB is set to 1 while the last block is being transferred, CMD52 is not issued. In this case, C52PUB is automatically cleared to 0 after the access end flag sets to 1.

When C52PUB and RWREQ are set to 1 in the CMD53 (multiple block) read sequence, the block transfer enters the read wait state between blocks and CMD52 is automatically issued. C52PUB is automatically cleared to 0 after reception of the response to CMD52 is completed. Additionally, if C52PUB is set to 1 while the last block is being transferred, CMD52 is not issued. In this case, C52PUB is automatically cleared to 0 after the access end flag sets to 1.

If C52PUB is set to 1 in the CMD53 (multiple block) read sequence, you must set RWREQ to 1 in addition to C52PUB.

Set SD_ARG before setting C52PUB to 1.

Set C52PUB to 1 after the response end flag sets.

48.2.21 SDIO_INFO1 : SDIO Interrupt Flag Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x06C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EXWT	EXPUB52	—	—	—	—	—	—	—	—	—	—	—	—	—	IOIRQ
Value after reset:	0 ^{*1}	0 ^{*1}	0	0	0	0	0	0	0	0	0	0	0	x	x	0 ^{*1}

Bit	Symbol	Function	R/W
0	IOIRQ	SDIO Interrupt Status Flag 0: No SDIO interrupt detected 1: SDIO interrupt detected	R/(W) ^{*2}
2:1	—	The read values are undefined. The write value should be 1.	R/W
13:3	—	These bits are read as 0. The write value should be 0.	R/W
14	EXPUB52	EXPUB52 Status Flag Indicates the status of the EXPUB52.	R/(W) ^{*2}
15	EXWT	EXWT Status Flag Indicates the status of the EXWT.	R/(W) ^{*2}

Bit	Symbol	Function	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The value is initialized by a reset and also on reset triggered by the `SOFT_RST.SDRST` flag.

Note 2. Only 0 can be written to clear the bit.

The `SDIO_INFO1` register indicates the status of the SDIO card access. Set flags to be cleared to 0. Set flags that are not being cleared to 1.

IOIRQ flag (SDIO Interrupt Status Flag)

The IOIRQ flag indicates that an SDIO interrupt occurred.

[Setting condition]

- When SDIO interrupt from an SDIO card is received while `INTEN` in `SDIO_MODE` is set to 1.

[Clearing condition]

- When 0 is written to IOIRQ.*¹

Note 1. Before clearing this bit, access the SDIO card to negate the SDIO interrupt signal from the SDIO card. If the interrupt signal is not negated, this bit can be set again.

EXPUB52 flag (EXPUB52 Status Flag)

The EXPUB52 flag indicates the EXPUB52 status.

[Setting conditions]

- While the last block in the CMD53 (multiple block) sequence is transferred, `C52PUB` in `SDIO_MODE` is set to 1.
- While `C52PUB` is set to 1 in the CMD53 (multiple block) write sequence, the last block is transferred.

[Clearing condition]

- When 0 is written to EXPUB52.

EXWT flag (EXWT Status Flag)

The EXWT flag indicates the EXWT status.

[Setting condition]

- While the last block in the CMD53 (multiple block) read sequence is transferred, `RWREQ` in `SDIO_MODE` is set to 1.

[Clearing condition]

- When 0 is written to EXWT.

48.2.22 SDIO_INFO1_MASK : SDIO INFO1 Interrupt Mask Register

Base address: $\text{SDHIn} = 0x4025_2000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{SDHIn_NS} = 0x5025_2000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x070

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EXWT M	EXPU B52M	—	—	—	—	—	—	—	—	—	—	—	—	—	IOIRQ M
Value after reset:	1	1	0	0	0	0	0	0	0	0	0	0	0	1	1	1

Bit	Symbol	Function	R/W
0	IOIRQM	IOIRQ Interrupt Mask Control 0: Do not mask IOIRQ interrupts 1: Mask IOIRQ interrupts	R/W
2:1	—	These bits are read as 1. The write value should be 1.	R/W
13:3	—	These bits are read as 0. The write value should be 0.	R/W
14	EXPUB52M	EXPUB52 Interrupt Request Mask Control 0: Do not mask EXPUB52 interrupt requests 1: Mask EXPUB52 interrupt requests	R/W
15	EXWTM	EXWT Interrupt Request Mask Control 0: Do not mask EXWT interrupt requests 1: Mask EXWT interrupt requests	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The SDIO_INFO1_MASK register enables or disables interrupt requests from the status flags in the SDIO_INFO1 register. See [Table 48.5](#) for details on the relationship between the status flags and the requested interrupt source.

48.2.23 SD_DMAEN : DMA Mode Enable Register

Base address: SDHln = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHln_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x1B0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DMAEN	—
Value after reset:	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	DMAEN	DMA Transfer Enable ^{*1 *2} 0: Disable use of DMA transfer to access SD_BUF0 register 1: Enable use of DMA transfer to access SD_BUF0 register	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	—	This bit is read as 1. The write value should be 1.	R/W
11:5	—	These bits are read as 0. The write value should be 0.	R/W
12	—	This bit is read as 1. The write value should be 1.	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Do not rewrite this bit when the SD_INFO2.CBSY bit is 1.

Note 2. When the SD_INFO2_MASK.BWEM bit is 0 or the SD_INFO2_MASK.BREM bit is 0, set the SD_DMAEN.DMAEN bit to 0. When the SD_DMAEN.DMAEN bit is 1, set the SD_INFO2_MASK.BWEM bit to 1 and the SD_INFO2_MASK.BREM bit to 1.

The SD_DMAEN register enables or disables DMA transfers.

DMAEN bit (DMA Transfer Enable)

When using DMA transfer to access the SD buffer, set the DMAEN bit to 1 before setting the SD_CMD register.

48.2.24 SOFT_RST : Software Reset Register

Base address: SDHIn = $0x4025_2000 + 0x0400 \times n$ ($n = 0, 1$)
 SDHIn_NS = $0x5025_2000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x1C0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SDRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

Bit	Symbol	Function	R/W
0	SDRST	Software Reset Control 0: Reset SD/MMC Host Interface software 1: Cancel reset of SD/MMC Host Interface software	R/W
2:1	—	These bits are read as 1. The write value should be 1.	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Table 48.4 lists the bits and flags initialized by SD/MMC Host Interface software reset.

Table 48.4 Bits and flags initialized by SD/MMC Host Interface software reset

Register	Bit/flag
SD_STOP	SEC, STP
SD_INFO1	RSPEND, ACEND
SD_INFO2	CMDE, CRCE, ENDE, DTO, ILW, ILR, RSPTO, BRE, BWE, SD_CLK_CTRLLEN, CBSY, ILA
SD_CLK_CTRL	CLKEN
SD_OPTION	CTOP[3:0], TOP[3:0], TOUTMASK, WIDTH8, WIDTH
SD_ERR_STS1	CMDE0, CMDE1, RSPLNE0, RSPLNE1, RDLENE, CRCLNE, RSPCRCE0, RSPCRCE1, RDCRCE, CRCTKE, CRCTK[2:0]
SD_ERR_STS2	RSPTO0, RSPTO1, BSYTO0, BSYTO1, RDTO, CRCTO, CRCBSYTO
SDIO_INFO1	IOIRQ, EXPUB52, EXWT
SDIF_MODE	NOCHKCR

48.2.25 SDIF_MODE : SD Interface Mode Setting Register

Base address: SDHIn = $0x4025_2000 + 0x0400 \times n$ ($n = 0, 1$)
 SDHIn_NS = $0x5025_2000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x1CC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	NOCHKCR	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0 ^{*1}	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	NOCHKCR	CRC Check Mask CRC check mask bit for MMC test commands. Set when CRC16 or CRC status value check is not executed. 0: Enable CRC check 1: Disable CRC Check (ignore CRC16 valued when reading and ignore CRC status value when writing)	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The value is initialized by a reset and also on reset triggered by the SOFT_RST.SDRST flag.

NOCHKCR bit (CRC Check Mask)

The NOCHKCR bit is used for MMC test commands. This bit is set when CRC16 or CRC status value check is not executed.

48.2.26 EXT_SWAP : Swap Control Register

Base address: SDHIn = 0x4025_2000 + 0x0400 × n (n = 0, 1)
SDHIn_NS = 0x5025_2000 + 0x0400 × n (n = 0, 1)

Offset address: 0x1E0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	BRSW P	BWS WP	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
6	BWSWP	SD_BUF0 Swap Write*1 0: Normal write operation 1: Swap the byte endian order before writing to SD_BUF0 register	R/W
7	BRSWP	SD_BUF0 Swap Read*1 0: Normal read operation 1: Swap the byte endian order before reading SD_BUF0 register	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Do not rewrite this bit when the SD_INFO2.CBSY flag is 1.

The EXT_SWAP register selects whether or not the byte endian order is swapped when accessing the SD_BUF0 register. See [section 48.3.1. SD/MMC Interface](#) for details on the differences in accessing the SD_BUF0 register based on the EXT_SWAP register value.

48.3 Operation

48.3.1 SD/MMC Interface

When data is read from the SD card/MMC, the process is as follows:

1. The SD/MMC Host Interface receives data from the SD card/MMC through the SDnDAT signal (see [Figure 48.2](#) and [Figure 48.3](#)).
2. The received data is stored in SD_BUF of the MMC Host Interface (see [Figure 48.4](#)).

3. The data stored in SD_BUF is read from SD_BUF0 (see [Figure 48.5](#)).

When data is written to the SD card/MMC, the specified procedure is reversed.

When accessing SD_BUF0, pay attention to the transfer order in SDnDAT and the store order in SD_BUF. If required, you can change the byte endian of the data read from or written to SD_BUF0 using the SDSWAP register. See [Figure 48.6](#).

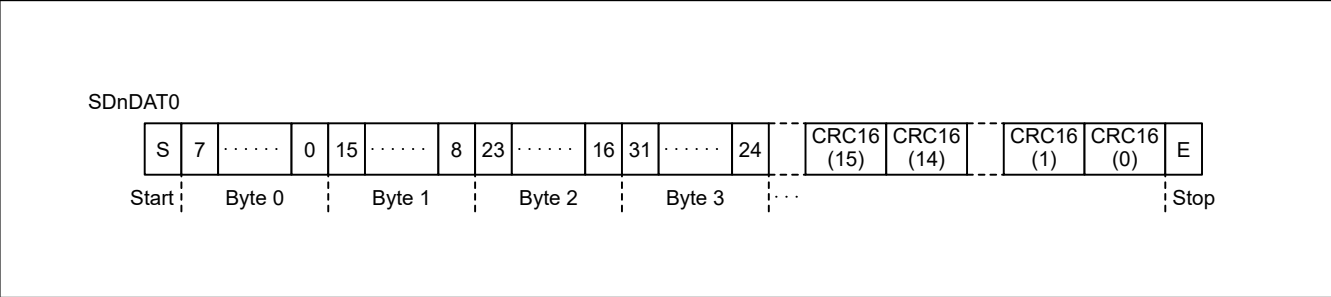


Figure 48.2 SDnDAT in 1-bit width mode

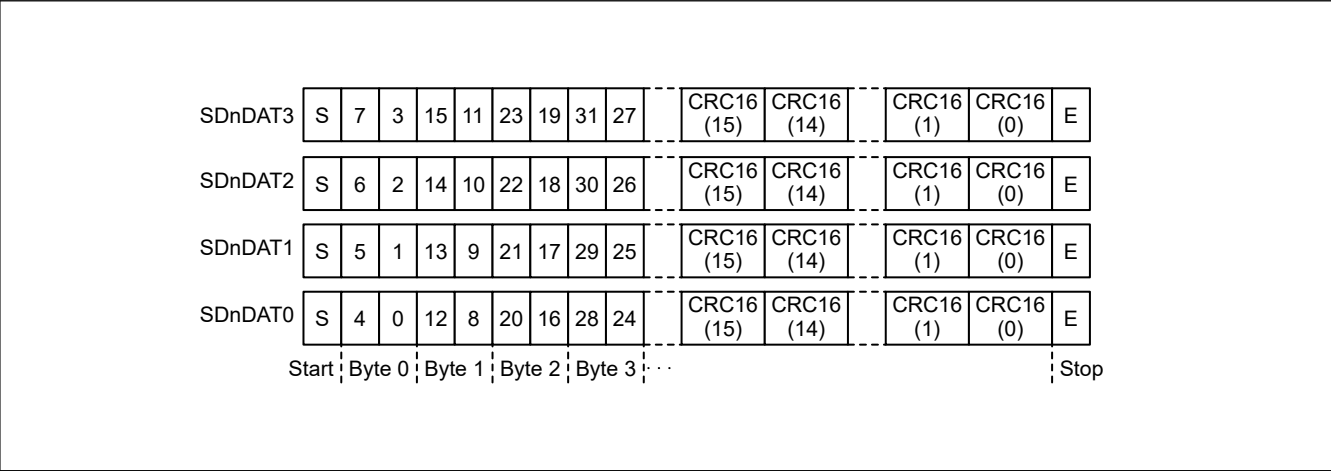


Figure 48.3 SDnDAT in 4-bit width mode

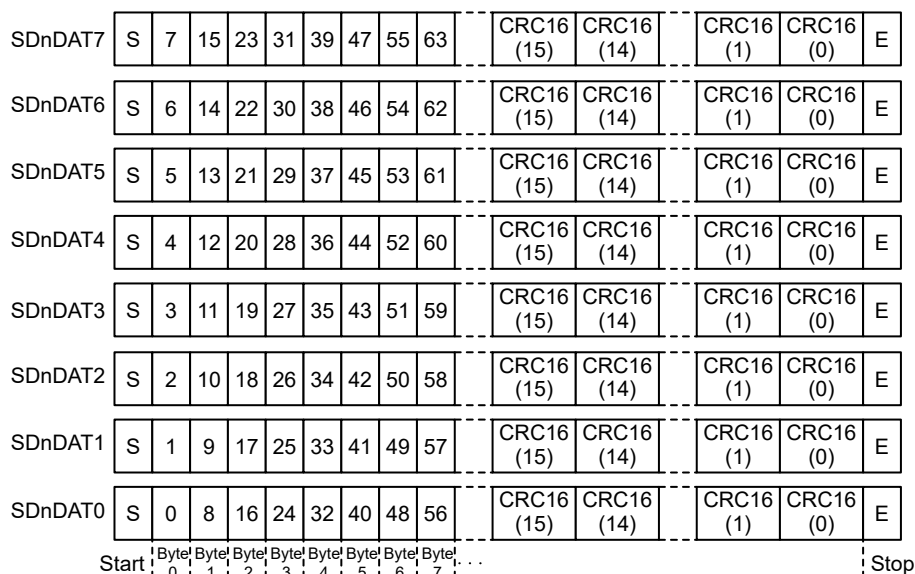


Figure 48.4 SDnDAT in 8-bit width mode

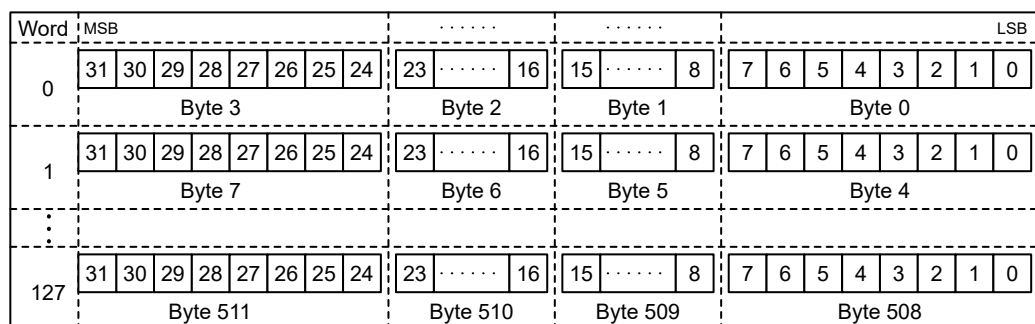
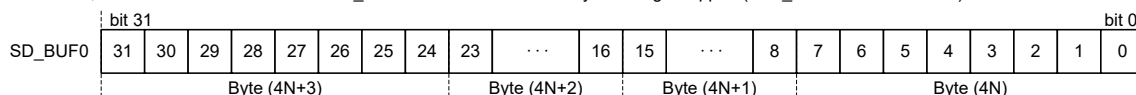


Figure 48.5 SD_BUF store data

- 32-bit access, in the case where the data in SD_BUF0 are read without the bytes being swapped (EXT_SWAP: BRSWAP = 0)



- 32-bit access, in the case where the data in SD_BUF0 are read with the bytes being swapped (EXT_SWAP: BRSWAP = 1)

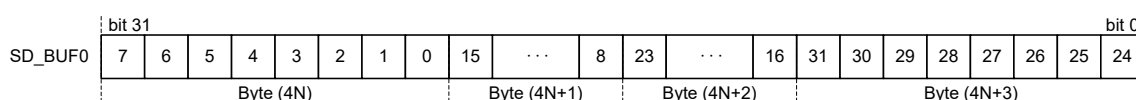


Figure 48.6 Read from SD_BUF0

48.3.2 Card Detect/Write Protect

48.3.2.1 Card Detect

The SD/MMC Host Interface has two types of card detect functions.

(1) Card detect with SDnCD (n = 0, 1)

Figure 48.7 shows the timing for card detect using SDnCD. SDnCD is connected to the card socket and pulled up on the host device. The resistance of the pull-up resistor is determined by the specification of the SD/MMC host device.

(2) Card insertion

SDnCD is pulled down when a card is inserted. At this point, if SDnCD is pulled down for the Mcycle period (set in SD_OPTION), SDCDIN in SD_INFO1 is set to 1. It is cleared by writing 0.

(3) Card removal

SDnCD is pulled up when a card is removed. At this point, if SDnCD is pulled up for the Mcycle period (set in SD_OPTION), SDCDRM in SD_INFO1 is set to 1. It is cleared by writing 0.

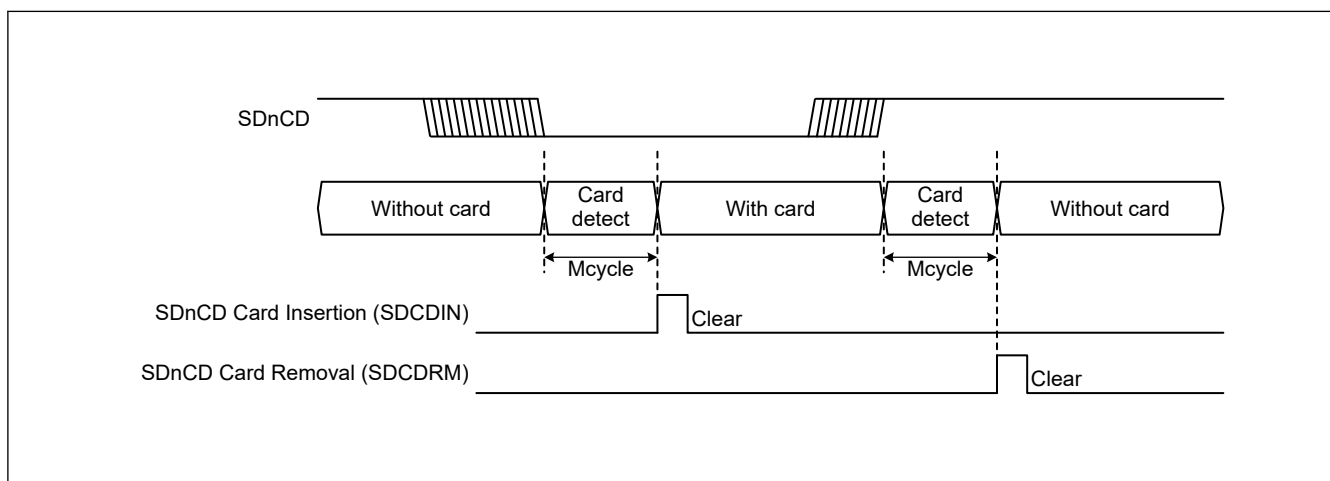


Figure 48.7 Example of card detect with SDnCD

(4) SD card detect with SDnDAT3 (n = 0, 1)

Figure 48.8 shows the timing when the SD card is detected with SDnDAT3. In addition, SDnDAT3 is pulled down by the host device, and the resistance value for pulling down is determined by the specification of the SD host device.

(5) Card insertion

When an SD card is inserted, SDnDAT3 is pulled up and SDD3IN in SD_INFO1 is set to 1. It is cleared by writing 0.

(6) Card removal

When an SD card is removed, SDnDAT3 is pulled down and SDD3RM in SD_INFO1 is set to 1. It is cleared by writing 0.

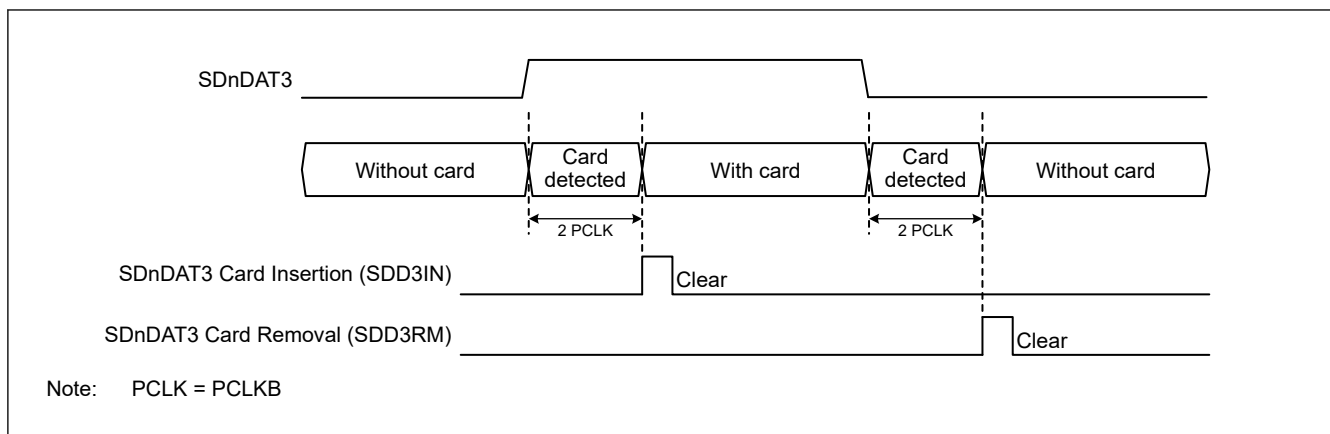


Figure 48.8 SD card detect with SDnDAT3

48.3.2.2 Write Protect

The SD/MMC Host Interface has two types of write protect functions.

(1) Write protect with SDnWP (n = 0, 1)

SDnWP is connected to the card socket and pulled up or pulled down by the card insertion. The selection of pulling up or pulling down and the resistance value is determined by the specification of the SD host device. When the SDnWP state is reflected to SDWPMON in SD_INFO1, the write protect state is set after the SD card is inserted.

(2) Write protect with command

The internal write protection of the card and the lock/unlock operation of the card are realized by the command.

48.3.3 Interrupt Request and DMA Transfer Request

48.3.3.1 Interrupts

Table 48.5 lists the SDHI interrupt sources. The SDHI requests an interrupt when:

- The status flags in registers SD_INFO1, SD_INFO2, and SDIO_INFO1 set to 1.
- The associated bits in the SD_INFO1_MASK, SD_INFO2_MASK, and SDIO_INFO1_MASK registers are 0.

When clearing the status flags in registers SD_INFO1, SD_INFO2, and SDIO_INFO1, write 0 to the status flags to be cleared and write 1 to the status flags that are not being cleared.

Table 48.5 Interrupt sources

Interrupt sources	Status flag register		Interrupt mask register		Interrupt name
	Register symbol	Bit symbol	Register symbol	Bit symbol	Ch n (n = 0, 1)
Card Access Interrupt (CACI)	SD_INFO1	ACEND	SD_INFO1_MASK	ACENDM	SDHI_MMChn_ACCS
		RSPEND		RSPENDM	
	SD_INFO2	ILA	SD_INFO2_MASK	ILAM	
		BWE		BWEM	
		BRE		BREM	
		RSPTO		RSPTOM	
		ILR		ILRM	
		ILW		ILWM	
		DTO		DTOM	
		ENDE		ENDEM	
		CRCE		CRCEM	
		CMDE		CMDEM	
SDIO Access Interrupt (SDACI)	SDIO_INFO1	EXWT	SDIO_INFO1_MASK	EXWTM	SDHI_MMChn_SDIO
		EXPUB52		EXPUB52M	
		IOIRQ		IOIRQM	
Card Detect Interrupt (CDETI)	SD_INFO1	SDD3IN	SD_INFO1_MASK	SDD3INM	SDHI_MMChn_CARD
		SDD3RM		SDD3RMM	
		SDCDIN		SDCDINM	
		SDCDRM		SDCDRMM	

48.3.3.2 DMA Transfer Requests (SDHI_MMChn_ODMSDBREQ, n = 0, 1)

The SD/MMC Host Interface has two types of DMA transfer requests.

(1) SD_BUF write DMA transfer request

- When the BWE bit in SD_INFO2 is set to 1 while the DMAEN bit in SD_DMAEN is set to 1, the SD_BUF write DMA transfer request is asserted.
- The SD_BUF write DMA transfer request is negated when the last data in one block (based on the transfer data size set in SD_SIZE) is transferred. The SD_BUF write DMA transfer request is also negated by clearing the SDRST bit in SOFT_RST to 0 or setting the STP bit in SD_STOP to 1. However, if a communications error or timeout occurs at the DMA transfer, the SD_BUF write DMA transfer request is not negated.
- The BWE bit in SD_INFO2 is cleared after transfer of the last data in one block following a request for writing to SD_BUF by DMA transfer.
- The number of DMA transfers must be $n \times$ one block. (n = integer, one block = the transfer data size set in SD_SIZE)
- When the IOABT bit in SDIO_MODE is set to 1, the SD_BUF write DMA transfer request is negated.
- The DMA transfer request is also negated by clearing the DMAEN bit to 0. However, the DMA transfer request is asserted again when the DMAEN bit is set to 1 before writing to SD_CMD.
- Because the BWE bit in SD_INFO2 is not cleared in response to setting the STP/IOABT bit, or to a communications error or timeout, clear the bit to 0 before issuing the next command. The next request to write to SD_BUF by DMA transfer is not issued while the BWE bit is set.

(2) SD_BUF read DMA transfer request

- When the BRE bit in SD_INFO2 is set to 1 while the DMAEN bit in the SD_DMAEN register is set to 1, the SD_BUF read DMA transfer request is asserted.

- The SD_BUF read DMA transfer request is negated when the last data in one block (based on the transfer data size set in SD_SIZE) is transferred. The SD_BUF read DMA transfer request is also negated by clearing the SDRST bit in SOFT_RST to 0 or setting the STP bit in SD_STOP to 1. However, if a communications error or timeout occurs at the DMA transfer, the SD_BUF read DMA transfer request is not negated.
- The BRE bit in SD_INFO2 is cleared after transfer of the last data in one block following a request to write to SD_BUF by DMA transfer.
- The number of DMA transfers must be $n \times$ one block. (n = integer, one block = the transfer data size set in SD_SIZE)
- When the IOABT bit in SDIO_MODE is set to 1, the SD_BUF read DMA transfer request is negated.
- The DMA transfer request is also negated by clearing the DMAEN bit to 0. However, the DMA transfer request is asserted again when the DMAEN bit is set to 1 before writing to SD_CMD.
- Because the BRE bit in SD_INFO2 is not cleared in response to setting the STP/IOABT bit or in response to a communications error or timeout, clear the bit to 0 before issuing the next command. The next request to write to SD_BUF by DMA transfer is not issued while the BRE bit is set.

48.3.4 Communication Errors and Timeouts

When a communication error or timeout error occurs, depending on the type of error, the associated status flag in the SD_INFO2 register sets to 1. Also, depending on the source of the error, the associated flag in the SD_ERR_STS1 or SD_ERR_STS2 register sets to 1.

The status flags in registers SD_ERR_STS1 and SD_ERR_STS2 clear to 0 by writing to the SD_CMD register, or by setting the SOFT_RST.SDRST bit to 0.

Table 48.6 Communication errors

Communication error	Interrupt flag register		Error status register		This occurs when...
	Register symbol	Bit symbol	Register symbol	Bit symbol	
End bit error	SD_INFO2	ENDE	SD_ERR_STS1	CRCLENE	The CRC status token length is in error
				RDLENE	The read data length is in error
				RSPLENE1	The response length is in error ^{*1}
				RSPLENE0	The response length is in error ^{*2}
CRC error		CRCE		CRCTKE	The CRC status token is in error
				RDCRCE	There is a CRC error in the read data
				RSPCRCE1	There is a CRC error in the response ^{*1}
				RSPCRCE0	There is a CRC error in the response ^{*2}
Command error		CMDE		CMDE1	The command index field value for the transmitted command and received response do not match ^{*1}
				CMDE0	The command index field value for the transmitted command and received response do not match ^{*2}

Note 1. CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD_CMD, CMD12 when the STP bit in SD_STOP is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO_MODE is set to 1.

Note 2. CMD other than CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD_CMD, CMD12 when the STP bit in SD_STOP is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO_MODE is set to 1.

Table 48.7 Timeouts

Timeout	Interrupt flag register		Error status register		This occurs when...	
	Register symbol	Bit symbol	Register symbol	Bit symbol		
Response timeout	SD_INFO2	RSPT0	SD_ERR_STS2	RSPT01	A response is not received even after a minimum of 640 SDHI clock cycles elapse ^{*1}	
				RSPT00	A response is not received even after a minimum of 640 SDHI clock cycles elapse ^{*2}	
Data timeout (excluding response timeout)		DTO		CRCBSYTO	After the CRC status token is received, the SDHI is busy for at least the period set ^{*3}	
				CRCTO	After the write data is transmitted, the CRC status token is not received even after at least the period set ^{*3} elapses	
					RDTO	After the read command is issued, the read data is not received even after at least the period set ^{*3} elapses
						After the read data is received, the next block read data is not received even after at least the period set ^{*3} elapses
					After the SDHI exits the read wait state, the next block read data is not received even after at least the period set ^{*3} elapses	
				BSYTO1	After CMD12 is issued during the command sequence, the SDHI is busy for at least the period set ^{*3}	
BSYTO0		After the R1b response is received, the SDHI is busy for at least the period set ^{*3} (a command other than CMD12 is issued during the command sequence)				

Note 1. CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD_CMD, CMD12 when the STP bit in SD_STOP is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO_MODE is set to 1.

Note 2. CMD other than CMD12 when automatic issuing is enabled for multiple block transfer by the setting in SD_CMD, CMD12 when the STP bit in SD_STOP bit is set to 1, or CMD52 when the C52PUB or IOABT bit in SDIO_MODE is set to 1.

Note 3. The period is set in the SD_OPTION.TOP[3:0] bits.

48.3.5 Command Without Data Transfer (SD/MMC)

The following legend is used for description of register read/write.

W (register name, value): Write to register

R (register name): Read from register

Figure 48.9 and Figure 48.10 show example flows.

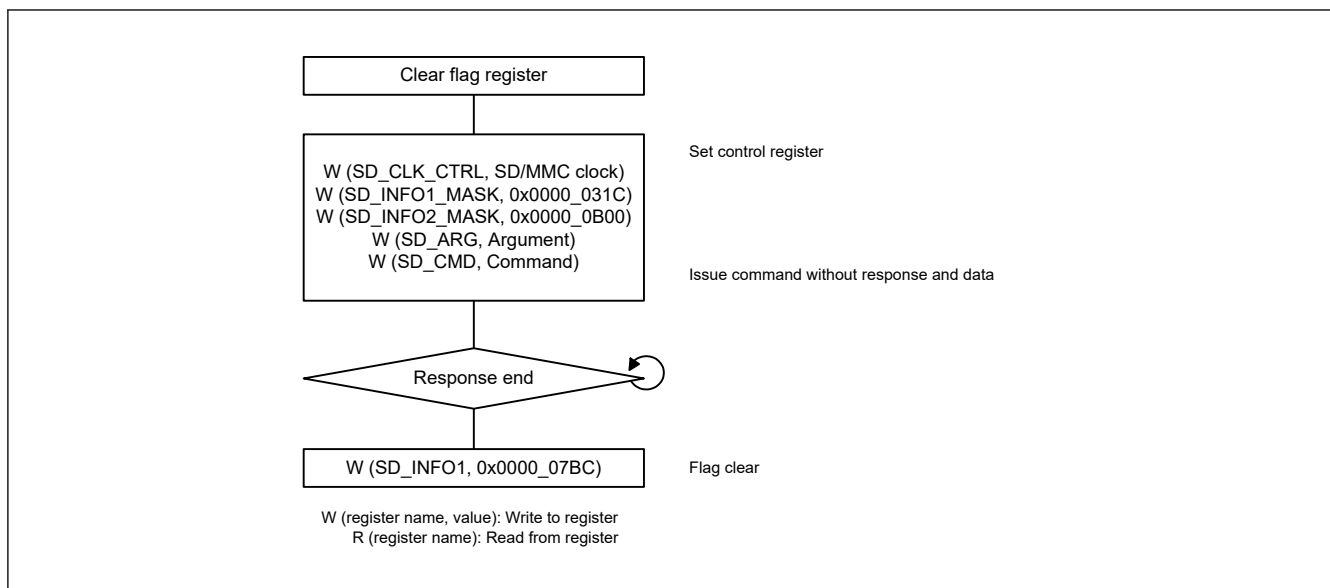


Figure 48.9 Example flow of command without response or data

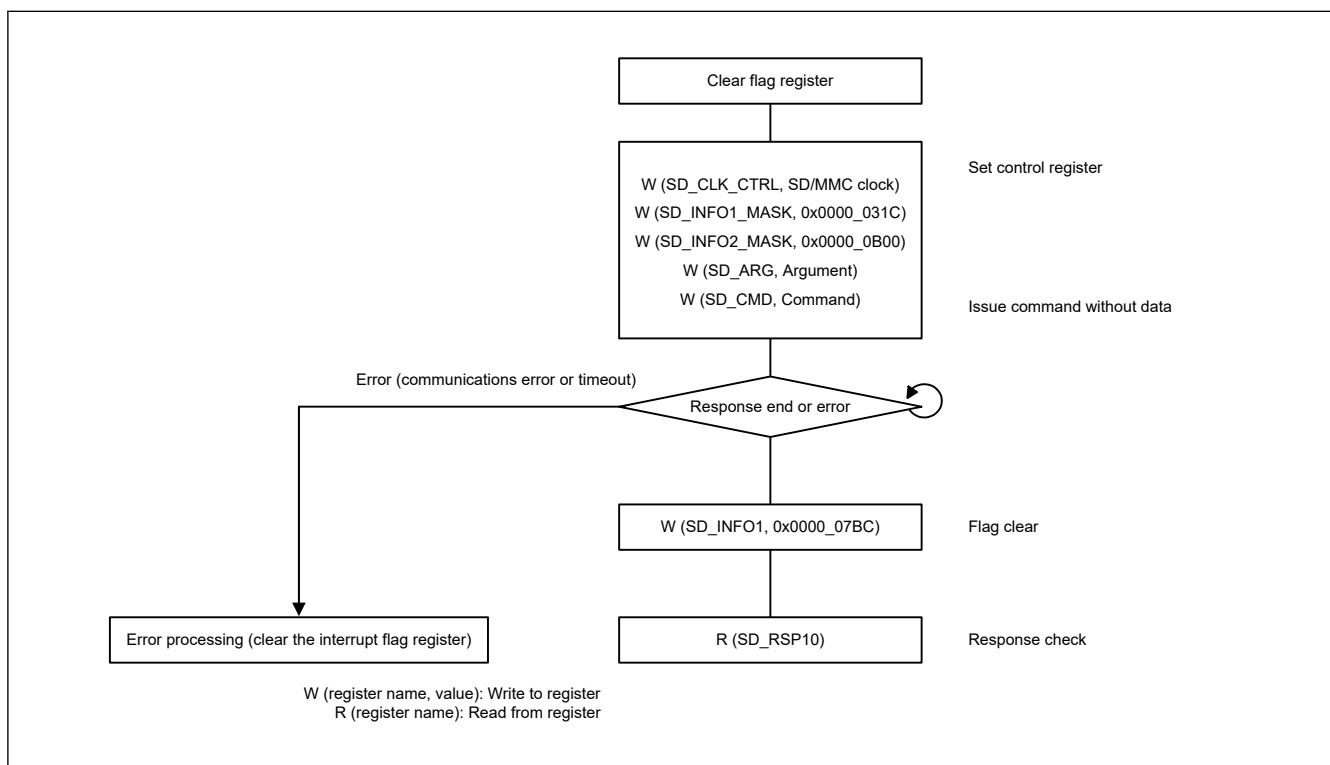


Figure 48.10 Example flow of command without data

48.3.5.1 Operation for Command Without Data Transfer

The operation is described in the following section.

(1) Command without response and data

- a. Flag register clear
First, clear the bits in the flag register. (SD_INFO1 and SD_INFO2)
- b. Control register set
Set the SD/MMC clock and interrupt masking. (SD_CLK_CTRL, SD_INFO1_MASK, and SD_INFO2_MASK)
- c. Command issue
Set CMD argument in SD_ARG and write to SD_CMD.

Accordingly, CMD is issued, and the operation is started.

d. Flag clear

When transmission of a command is completed, RSPEND (response end) in SD_INFO1 is set to 1 to generate an interrupt. Clear RSPEND to 0.

(2) Command without data

a. Flag register clear

First, clear the bits in the flag register. (SD_INFO1 and SD_INFO2)

b. Control register set

Set the SD/MMC clock and interrupt masking. (SD_CLK_CTRL, SD_INFO1_MASK, and SD_INFO2_MASK)

c. Command issue

Set CMD argument in SD_ARG and write to the SD_CMD.
Accordingly, CMD is issued, and the operation is started.

d. Flag clear

When a response is received, RSPEND (response end) in SD_INFO1 is set to 1 to generate an interrupt. Clear RSPEND to 0.

e. Read a response from SD_RSP10. Additionally, perform error processing (clear the interrupt flag register) if a communication error or timeout occurs.

48.3.6 Single Block Read (SD/MMC)

Figure 48.11 shows an example flow of a single block read operation.

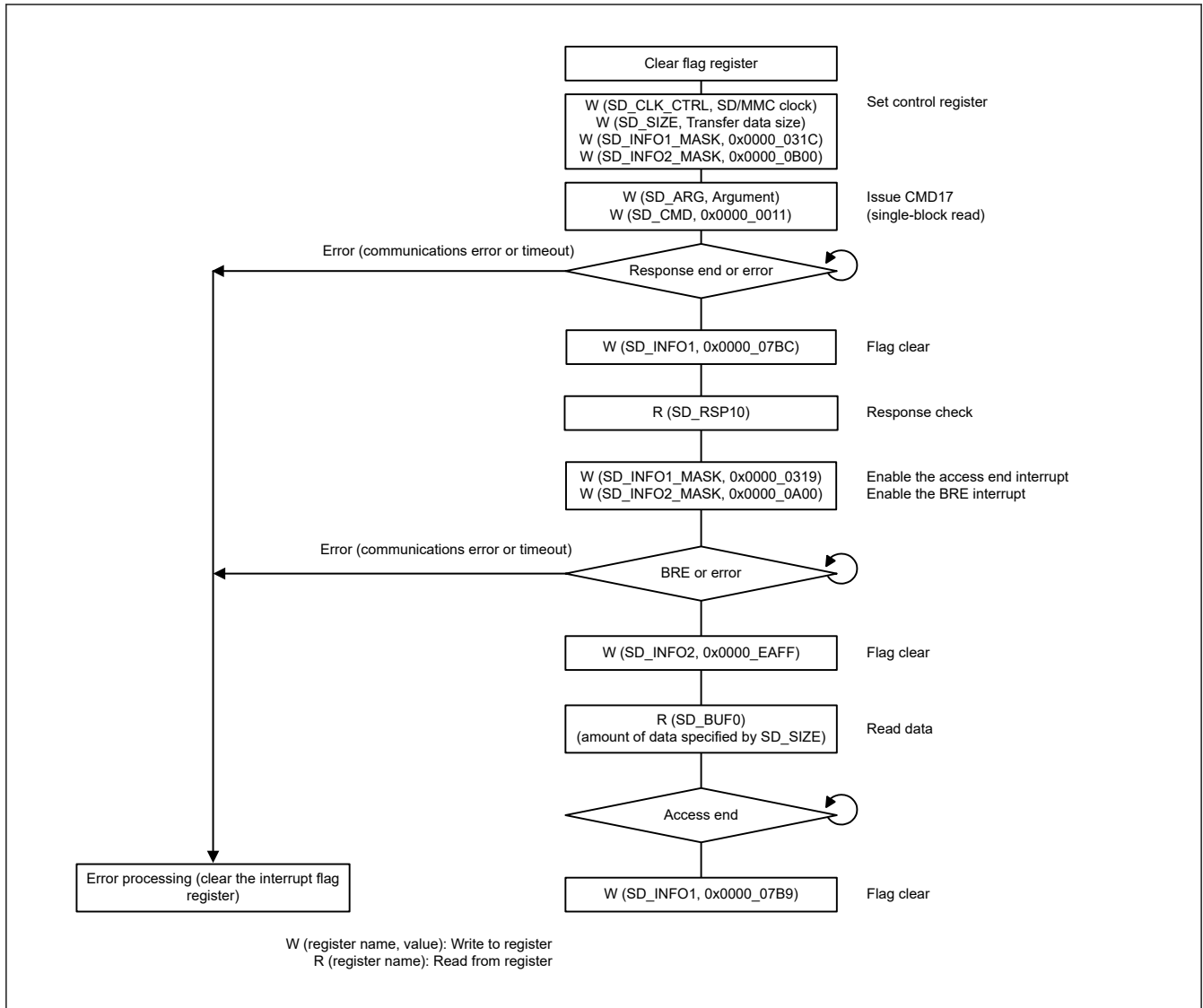


Figure 48.11 Example flow of single block read operation

48.3.6.1 Single Block Read Operation

The operation of the single block read is described as follows:

- Flag register clear**
First, clear the bits in the flag register (SD_INFO1 and SD_INFO2).
- Control register set**
Set the SD/MMC clock, transfer data size, interrupt mask (SD_CLK_CTRL, SD_SIZE, SD_INFO1_MASK, and SD_INFO2_MASK).
- Command issue (CMD17)**
Set CMD17 argument in SD_ARG and write 0x0000_0011 to SD_CMD. CMD17 is issued and the single block read operation is started.
- Response check**
On receiving the response, RSPEND (response end) in SD_INFO1 is set to 1 to generate an interrupt. Clear RSPEND to 0 and read the response from SD_RSP10. If the result of response decoding is an error, the command sequence can be halted by setting the STP bit in SD_STP or the IOABT bit in SDIO_MODE to 1. In addition, this causes CMD12 and CMD52 to not be issued. If the ACEND bit (access end) in SD_INFO1 is set, halting the command sequence also leads to the generation of an interrupt.
- Data receive from SD card/MMC and data read**

Write 0x0000_0319 to SD_INFO1_MASK to enable the access end interrupt. In addition, write 0x0000_0A00 to SD_INFO2_MASK to enable the BRE interrupt. When the data received from the SD card/MMC is completed, the BRE bit in SD_INFO2 is set to 1 to generate an interrupt. Clear the BRE bit to 0 and read the amount of data specified in SD_SIZE from SD_BUF0.

A communication error or timeout might be generated if data is being received while reading of SD_BUF0 is in progress.

f. Operation complete

When the data read from SD_BUF0 is completed, ACEND (access end) in SD_INFO1 is set to 1 to generate an interrupt. Clear ACEND to 0 to end the single block read operation.

Additionally, perform error processing (clear the interrupt flag register) if a communication error or timeout occurs.

48.3.7 Single Block Write (SD/MMC)

Figure 48.12 shows an example flow of a single block write operation.

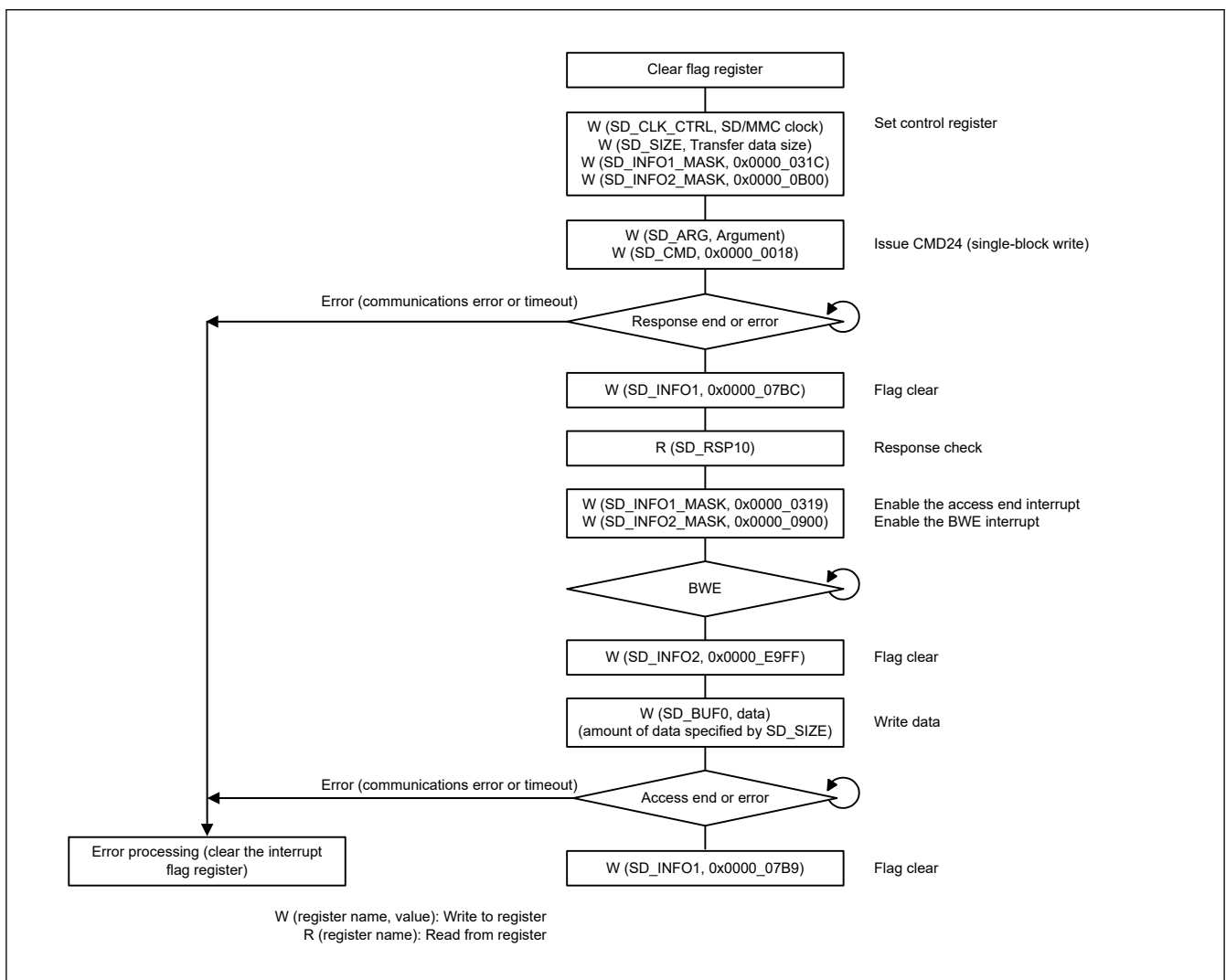


Figure 48.12 Example of single block write operation

48.3.7.1 Single Block Write Operation

The operation of the single block write is described as follows:

- Flag register clear
First, clear the bits in the flag register (SD_INFO1 and SD_INFO2).
- Control register set

Set the SD/MMC clock, transfer data size, interrupt mask (SD_CLK_CTRL, SD_SIZE, SD_INFO1_MASK, and SD_INFO2_MASK).

c. Command issue (CMD24)

Set CMD24 argument in SD_ARG and write 0x0000_0018 to SD_CMD. CMD24 is issued and the single block write operation is started.

d. Response check

On receiving the response, RSPEND (response end) in SD_INFO1 is set to 1 to generate an interrupt. Clear RSPEND to 0 and read the response from SD_RSP10. If the result of response decoding is an error, the command sequence can be halted by setting the STP bit in SD_STP or the IOABT bit in SDIO_MODE to 1. In addition, this causes CMD12 and CMD52 to not be issued. If the ACEND bit (access end) in SD_INFO is set, halting the command sequence also leads to the generation of an interrupt.

e. Data write and data transmit to SD card/MMC

Write 0x0000_0319 to SD_INFO1_MASK to enable the access end interrupt. In addition, write 0x0000_0900 to SD_INFO2_MASK to enable the BWE interrupt. When SD_BUF0 is ready for the data to be written, the BWE bit in SD_INFO2 is set to 1 to generate an interrupt. Clear the BWE bit to 0 and write the amount of data specified in SD_SIZE to SD_BUF0. When the data write to SD_BUF0 is completed, data is transmitted to the SD card. Then, the CRC status and busy state are received from the SD card/MMC.

However, a communications error or timeout might be generated if data is being transmitted after writing to SD_BUF0.

f. Operation complete

When the CRC status and busy state are received from the SD card/MMC, ACEND (access end) in SD_INFO1 is set to 1 to generate an interrupt. Clear the ACEND bit to 0 to end the single block write operation.

In addition, perform error processing (clear the interrupt flag register) if a communication error or timeout occurs.

48.3.8 Multiple Block Read (SD/MMC)

Figure 48.13 shows an example flow of a multiple block read operation.

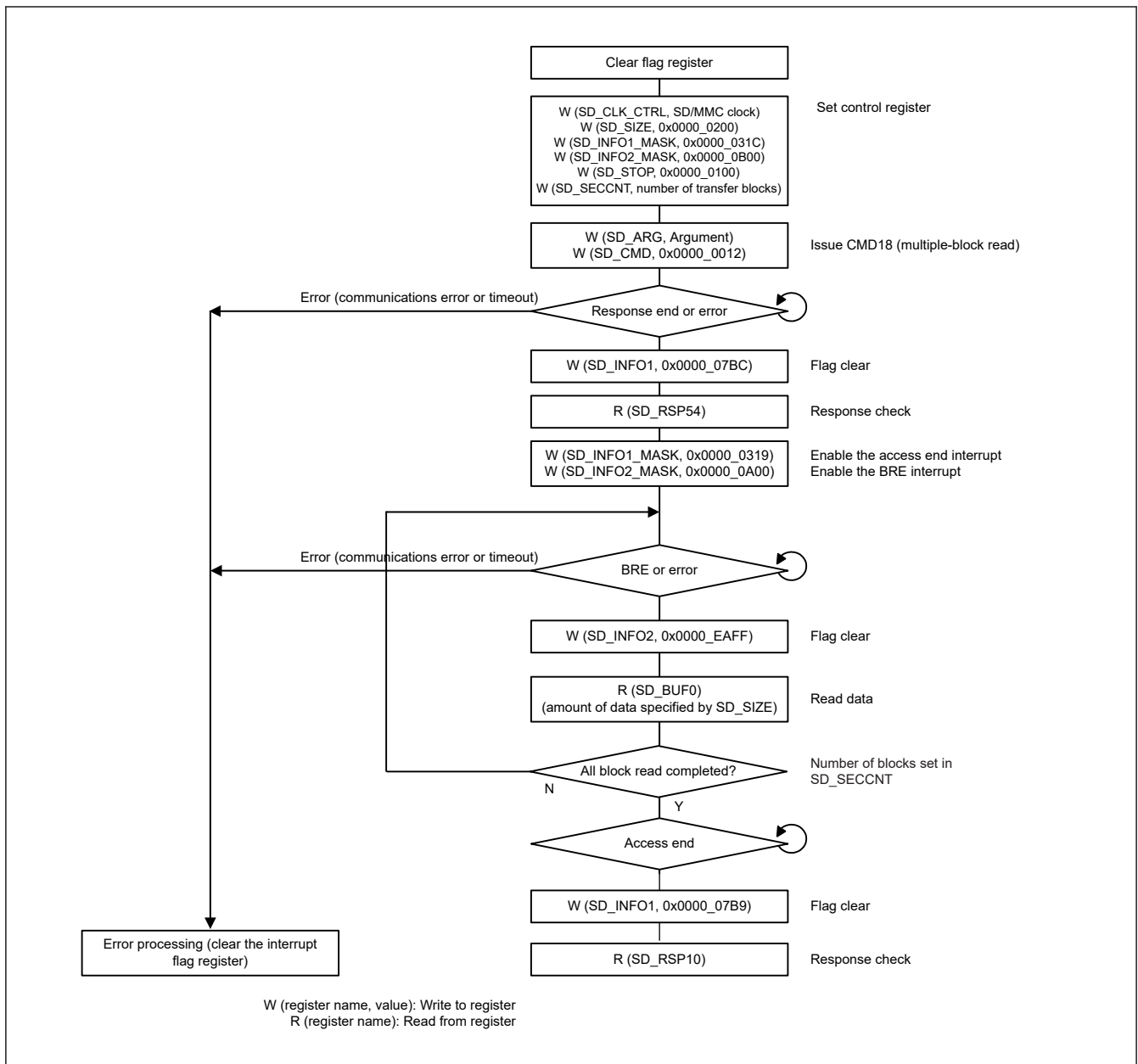


Figure 48.13 Example of multiple block read operation

48.3.8.1 Multiple Block Read Operation

The operation of the multiple block read is described as follows:

- Flag register clear**
First, clear the bits in the flag register (SD_INFO1 and SD_INFO2).
- Control register set**
Set the SD/MMC clock, transfer data size, interrupt mask (SD_CLK_CTRL, SD_SIZE, SD_INFO1_MASK, and SD_INFO2_MASK).
Set SEC in SD_STOP to 1, and set the number of transfer blocks in SD_SECCNT.
- Command issue (CMD18)**
Set CMD18 argument in SD_ARG and write 0x0000_0012 to SD_CMD. CMD18 is issued and the multiple block read operation is started.
- Response check**
On receiving the response, RSPEND (response end) in SD_INFO1 is set to 1 to generate an interrupt. Clear RSPEND to 0 and read the response from SD_RSP54. If the result of response decoding is an error, the command sequence can be

halted by setting the STP bit in SD_STP to 1. Setting the STP bit to 1 also causes CMD12 to be issued and the response received. If the command sequence is halted because the access end interrupt is enabled, an interrupt is generated when the ACEND bit (access end) bit in SD_INFO1 sets to 1 on completion of response reception. Clear the ACEND bit to 0 and read the response.

e. Data receive from SD card/MMC and data read

Write 0x0000_0319 to SD_INFO1_MASK to enable the access end interrupt. In addition, write 0x0000_0A00 to SD_INFO2_MASK to enable the BRE interrupt. When one-block data received from the SD card/MMC is completed, the BRE bit in SD_INFO2 is set to 1 to generate an interrupt. Clear the BRE bit to 0 and read the amount of data specified in SD_SIZE from SD_BUF0. Doing this repeats transfer of the number of blocks set in SD_SECCNT. However, a communication error or timeout might be generated if data is being received while reading of SD_BUF0 is in progress. CMD12 is automatically issued to stop multiblock transfer with the number of blocks that is set to SD_SECCNT and the response is received. At this point, CMD12 argument is automatically set to 0x0000_0000.

f. Operation complete

When all-block data read and the CMD12 response received are completed, ACEND (access end) in SD_INFO1 is set to 1 to generate an interrupt. Clear ACEND to 0 to read the response. This is the end of multiple block read operation. In addition, perform error processing (clear the interrupt flag register) if a communication error or timeout occurs.

48.3.9 Multiple Block Write (SD/MMC Using Internal Timer)

Figure 48.14 shows an example flow of a multiple block write using internal timer.

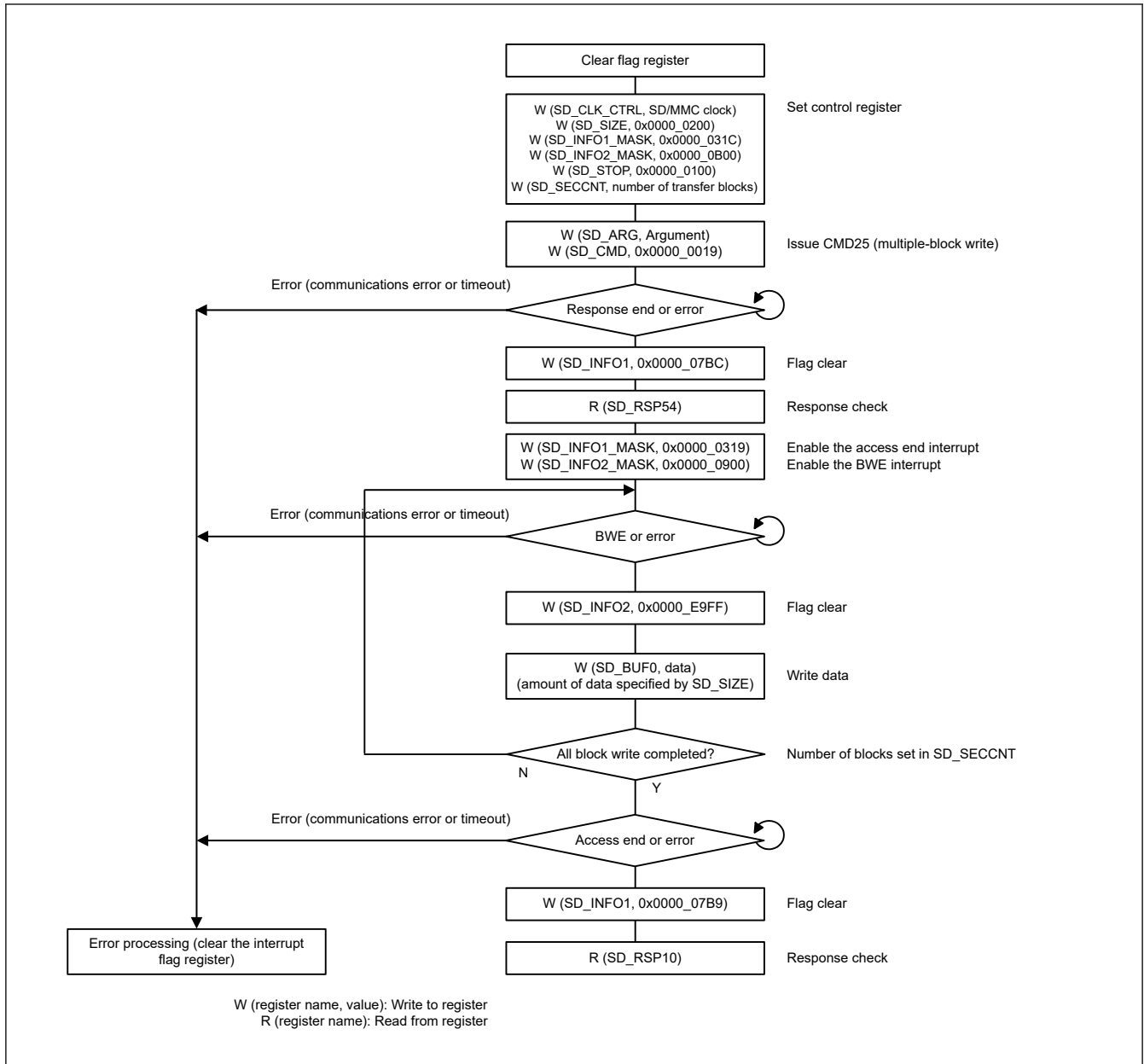


Figure 48.14 Example of multiple block write operation using internal timer

48.3.9.1 Multiple Block Write Operation Using Internal Timer

The operation of the multiple block write is described as follows:

- Flag register clear**
First, clear the bits in the flag register (SD_INFO1 and SD_INFO2).
- Control register set**
Set the SD/MMC clock, transfer data size, interrupt mask (SD_CLK_CTRL, SD_SIZE, SD_INFO1_MASK, and SD_INFO2_MASK).
Set the SEC bit in SD_STOP to 1, and set the number of transfer blocks in SD_SECCNT.
- Command issue (CMD25)**
Set CMD25 argument in SD_ARG and write 0x0000_0019 to SD_CMD. CMD25 is issued and the multiple block write operation is started.
- Response check**
On receiving the response, the RSPEND bit (response end) in SD_INFO1 is set to 1 to generate an interrupt.

Clear the RSPEND bit to 0 and read the response from SD_RSP54. If the result of response decoding is an error, the command sequence can be halted by setting the STP bit in SD_STP to 1. Setting the STP bit to 1 also causes CMD12 to be issued and the response received. If the command sequence is halted because the access end interrupt is enabled, an interrupt is generated by when the ACEND bit (access end) bit in SD_INFO1 sets to 1 on completion of response reception. Clear the ACEND bit to 0 and read the response.

e. Data write and data transmit to SD card/MMC

Write 0x0000_0319 to SD_INFO1_MASK to enable the access end interrupt. In addition, write 0x0000_0900 to SD_INFO2_MASK to enable the BWE interrupt. When SD_BUF0 is ready for the data to be written, the BWE bit in the SD_INFO2 register is set to 1 to generate an interrupt. Clear the BWE bit to 0 and write the amount of data specified in SD_SIZE to SD_BUF0. When the data write to SD_BUF0 is completed, data is transmitted to the SD card/MMC. The CRC status and busy state are received from the SD card/MMC. This repeats transfer of the number of blocks set in SD_SECCNT. However, a communication error or timeout might be generated if data is being received while writing to SD_BUF0 is in progress. CMD12 is automatically issued to stop multiblock transfer with the number of blocks which is set to SD_SECCNT and the response is received. At this point, CMD12 argument is automatically set to 0x0000_0000.

f. Operation complete

When all-block data transmit and the CRC status receive are completed, the ACEND bit (access end) in SD_INFO1 is set to 1 to generate an interrupt. Clear the ACEND bit to 0 to read the response. This is the end of multiple block write operation. Additionally, perform error processing (clear the interrupt flag register) if a communications error or timeout occurs.

48.3.10 Multiple Block Write (MMC Using External Timer)

Figure 48.15 shows an example flow of a multiple block write using an external timer.

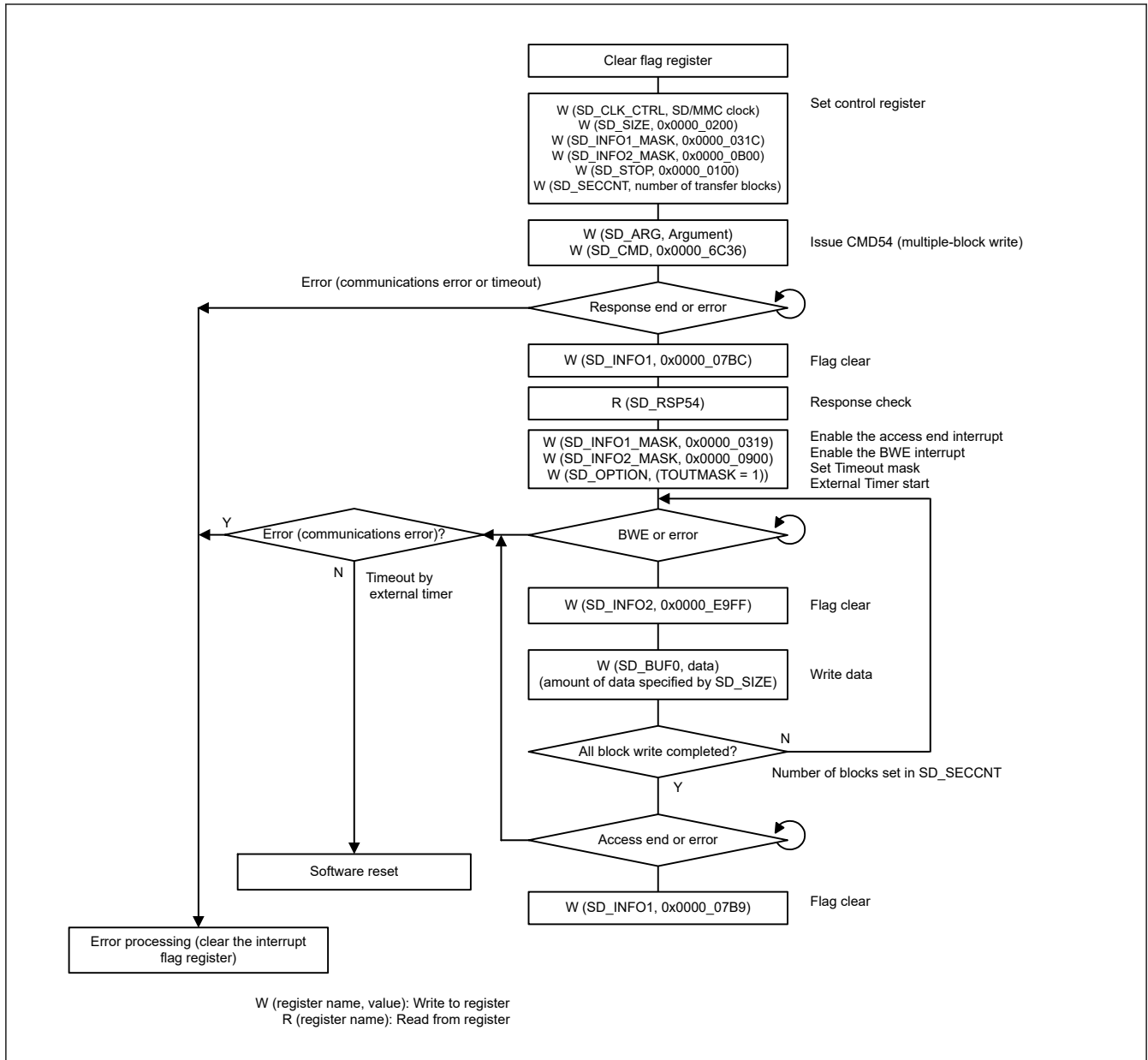


Figure 48.15 Example of multiple block write operation using external timer

48.3.10.1 Multiple Block Write Operation Using External Timer

The operation of the multiple block write is described as follows:

- Flag register clear**
First, clear the bits in the flag register (SD_INFO1 and SD_INFO2).
- Control register set**
Set the MMC clock, transfer data size, interrupt mask (SD_CLK_CTRL, SD_SIZE, SD_INFO1_MASK, and SD_INFO2_MASK).
Set the SEC bit in SD_STOP to 1, and set the number of transfer blocks in SD_SECCNT.
- Command issue (CMD54)**
Set CMD54 Argument in SD_ARG and write 0x0000_6C36 to SD_CMD. CMD54 is issued and the multiple block write operation is started.
- Response check**
On receiving the response, the RSPEND bit (response end) in SD_INFO1 is set to 1 to generate an interrupt. Clear the RSPEND bit to 0 and read the response from SD_RSP54. If the result of response decoding is an error, the command sequence can be halted by setting the STP bit in SD_STP to 1. Setting the STP bit to 1 also causes CMD12 to be issued

and the response received. If the command sequence is halted because the access end interrupt is enabled, an interrupt is generated by when the ACEND bit (access end) bit in SD_INFO1 sets to 1 on completion of response reception. Clear the ACEND bit to 0 and read the response.

e. Data write and data transmit to MMC

Write 0x0000_0319 to SD_INFO1_MASK to enable the access end interrupt, write 0x0000_0900 to SD_INFO2_MASK to enable the BWE interrupt and set 1 to TOUTMASK of SD_OPTION to inactivate timeout. In addition, start external timer. When SD_BUF0 is ready for the data to be written, the BWE bit in the SD_INFO2 register is set to 1 to generate an interrupt. Clear the BWE bit to 0 and write the amount of data specified in SD_SIZE to SD_BUF0. When the data write to SD_BUF0 is completed, data is transmitted to the MMC. The CRC status and busy state are received from the MMC. Doing this repeats transfer of the number of blocks set in SD_SECCNT. However, a communication error or timeout might be generated if data is being received while writing to SD_BUF0 is in progress.

f. Operation complete

When all-block data transmit and the CRC status receive are completed, the ACEND bit (access end) in SD_INFO1 is set to 1 to generate an interrupt. Clear the ACEND bit to 0 to read the response. This is the end of multiple block write operation. Additionally, perform error processing (clear the interrupt flag register) if a communications error or timeout occurs when receiving response. Execute software reset if a timeout by external timer occurs when transmitting data.

48.3.11 IO_RW_DIRECT Command (SD: CMD52)

Figure 48.16 shows an example flow of an IO_DIRECT command (CMD52) operation.

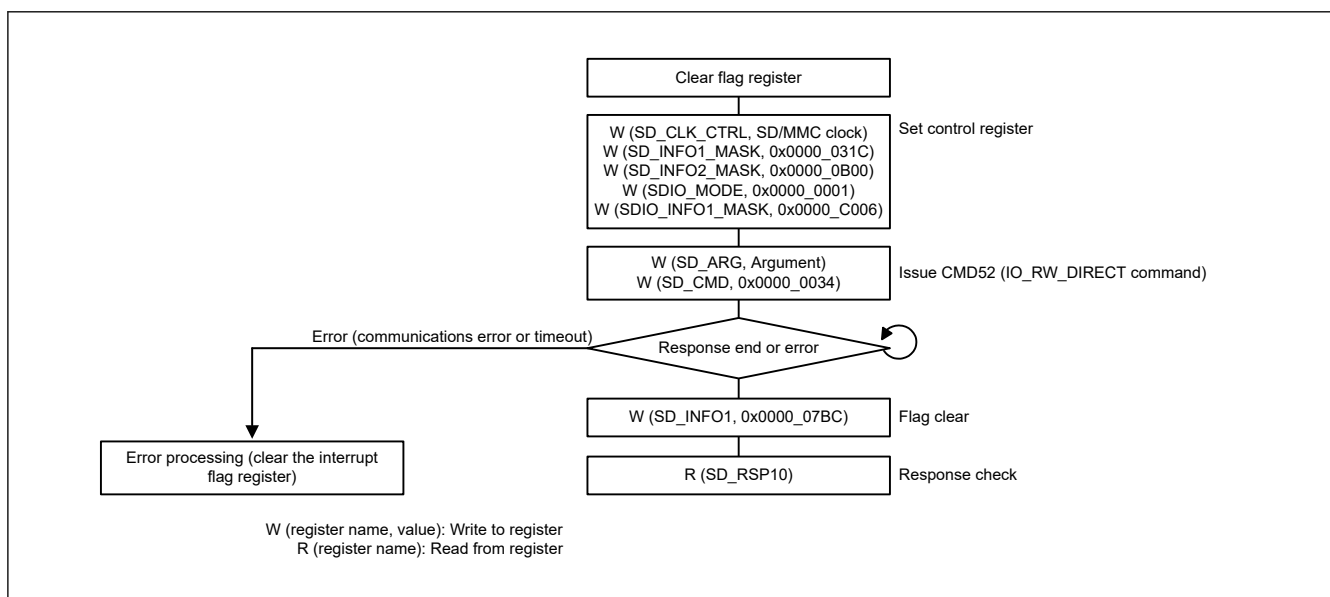


Figure 48.16 Example of IO_RW_DIRECT command (CMD52) operation

48.3.12 IO_RW_EXTENDED Command (SD: CMD53/Multiple Block Read)

Figure 48.17 shows an example flow for a CMD53 multiple block read operation.

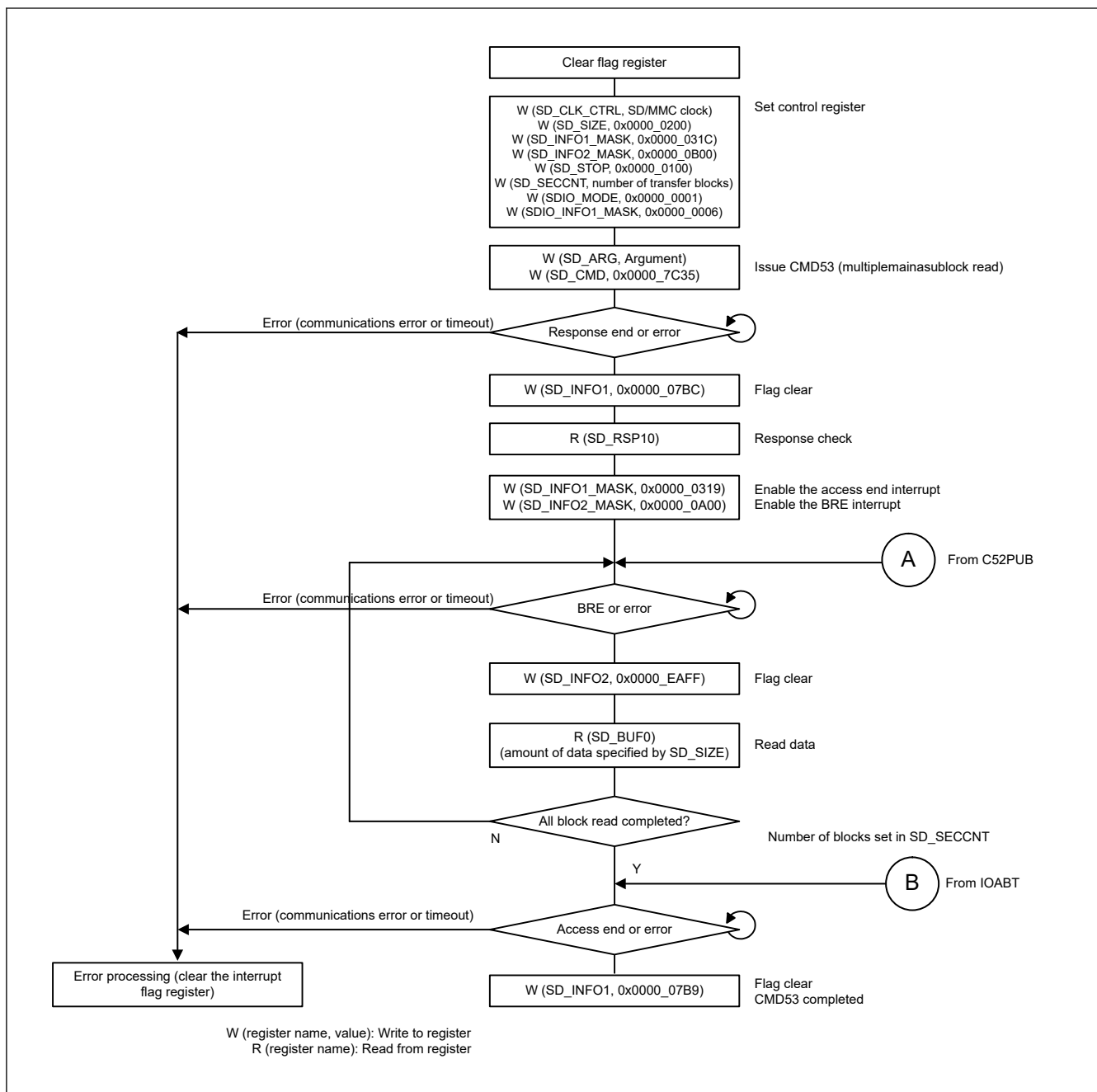


Figure 48.17 Example of IO_RW_EXTENDED command (CMD53) for multiple block read operation

Figure 48.18 shows an example flow when CMD52 (SDIO abort) is issued during a CMD53 multiple block read.

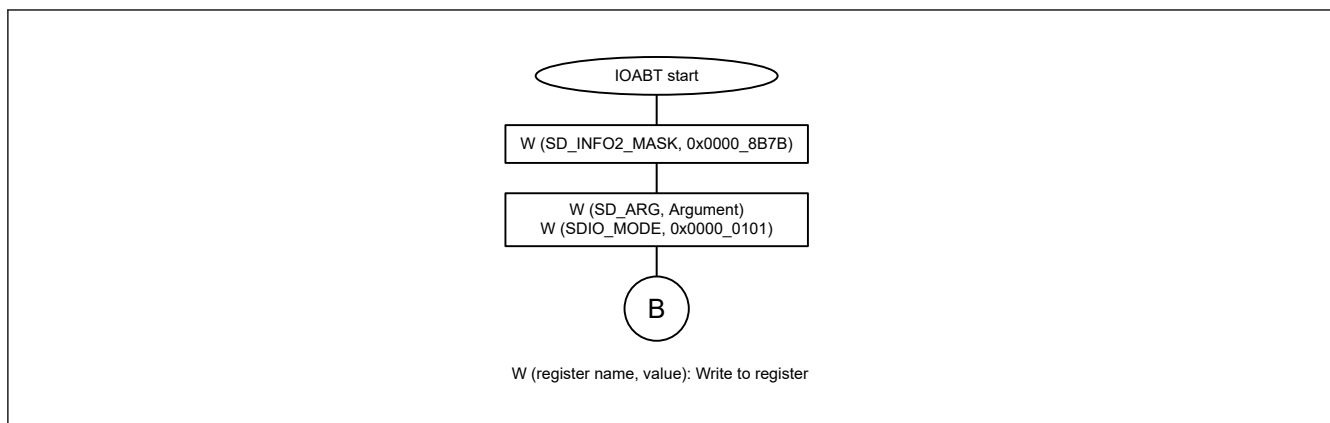


Figure 48.18 Flow when CMD52 (SDIO abort) is issued during a CMD53 multiple block read

Figure 48.19 shows an example flow when CMD52 (SDIO none abort) is issued at a CMD53 multiple block read while the SDHI is in the read wait state.

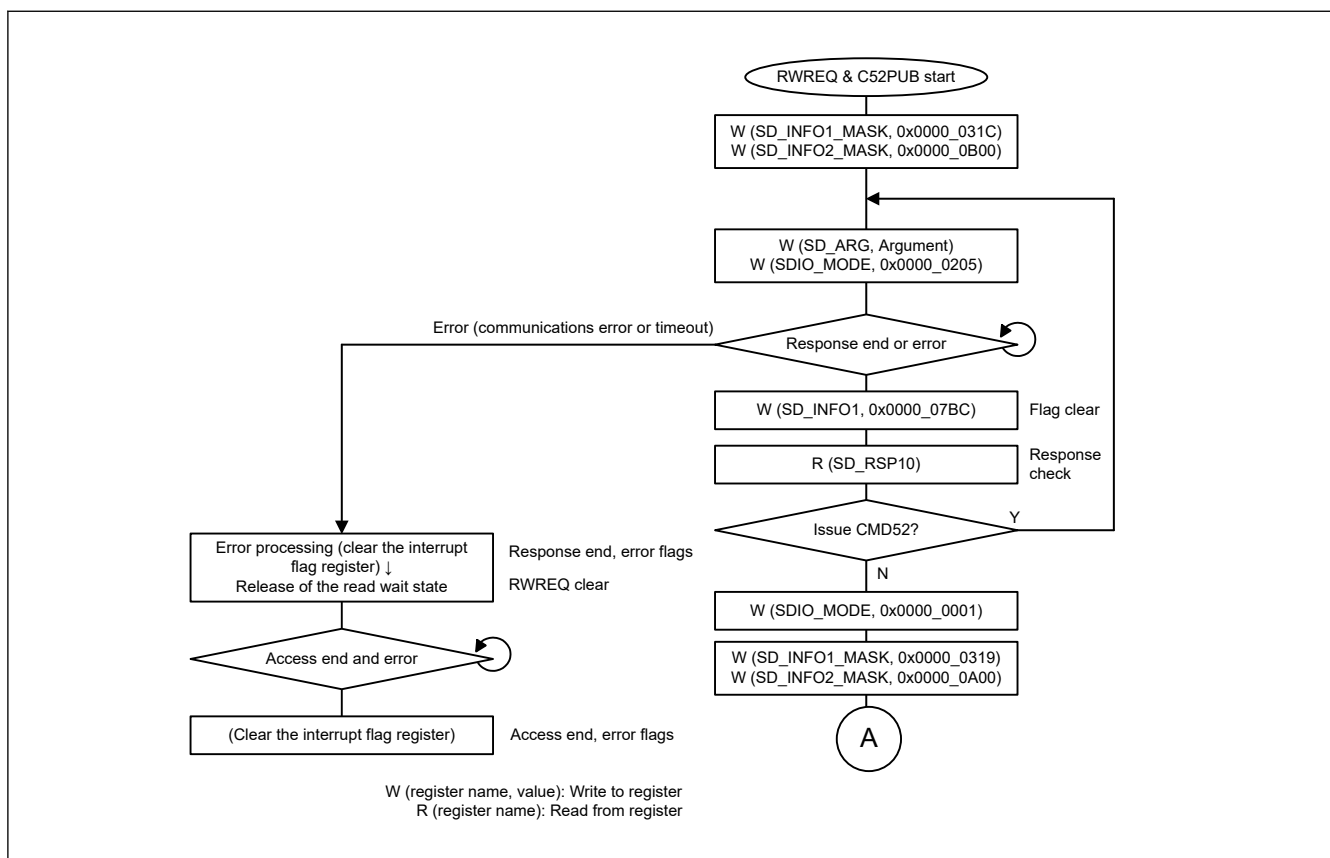


Figure 48.19 Flow when CMD52 (SDIO no abort) is issued during a CMD53 multiple block read while the SD Host Interface is in read wait state

48.3.13 IO_RW_EXTENDED Command (SD: CMD53/Multiple Block Write)

Figure 48.20 shows an example flow for a CMD53 multiple block write.

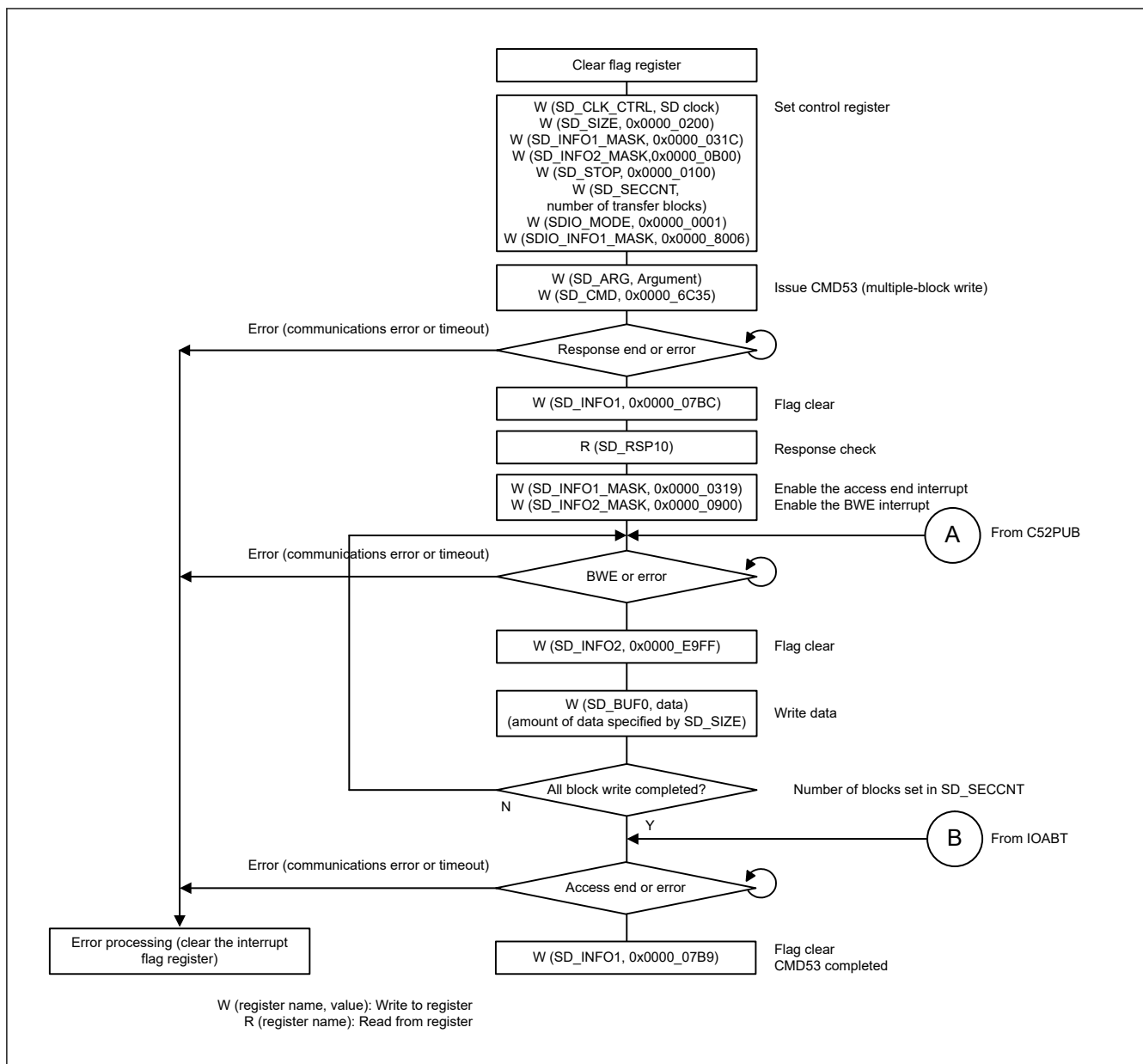


Figure 48.20 Example of IO_RW_EXTENDED command during a CMD53 multiple block write operation

Figure 48.21 shows an example flow when CMD52 (SDIO abort) is issued at CMD53 multiple block write.

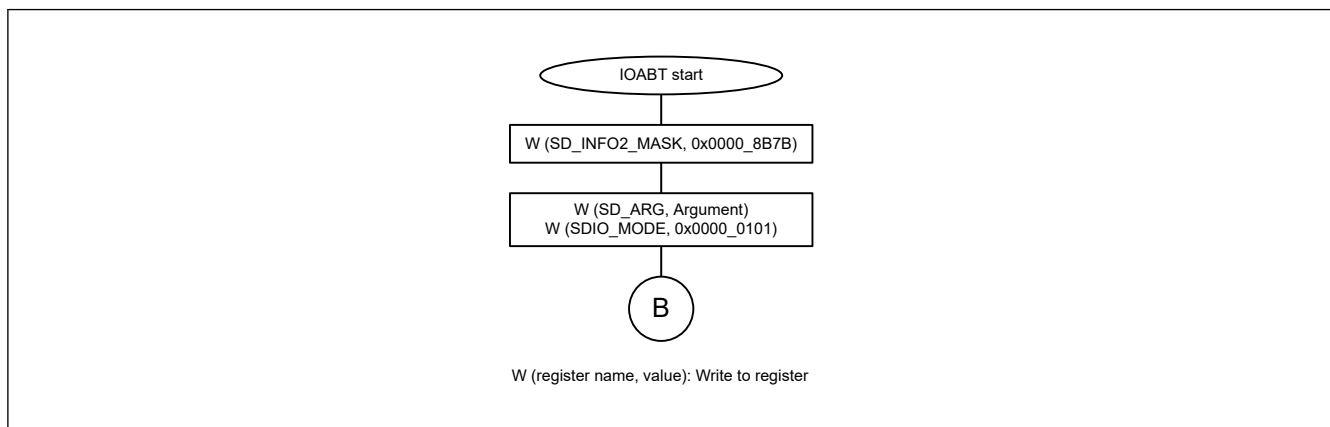


Figure 48.21 Flow when CMD52 (SDIO Abort) is issued during a CMD53 multiple block write

Figure 48.22 shows an example flow when CMD52 (SDIO none abort) is issued at CMD53 multiple block write.

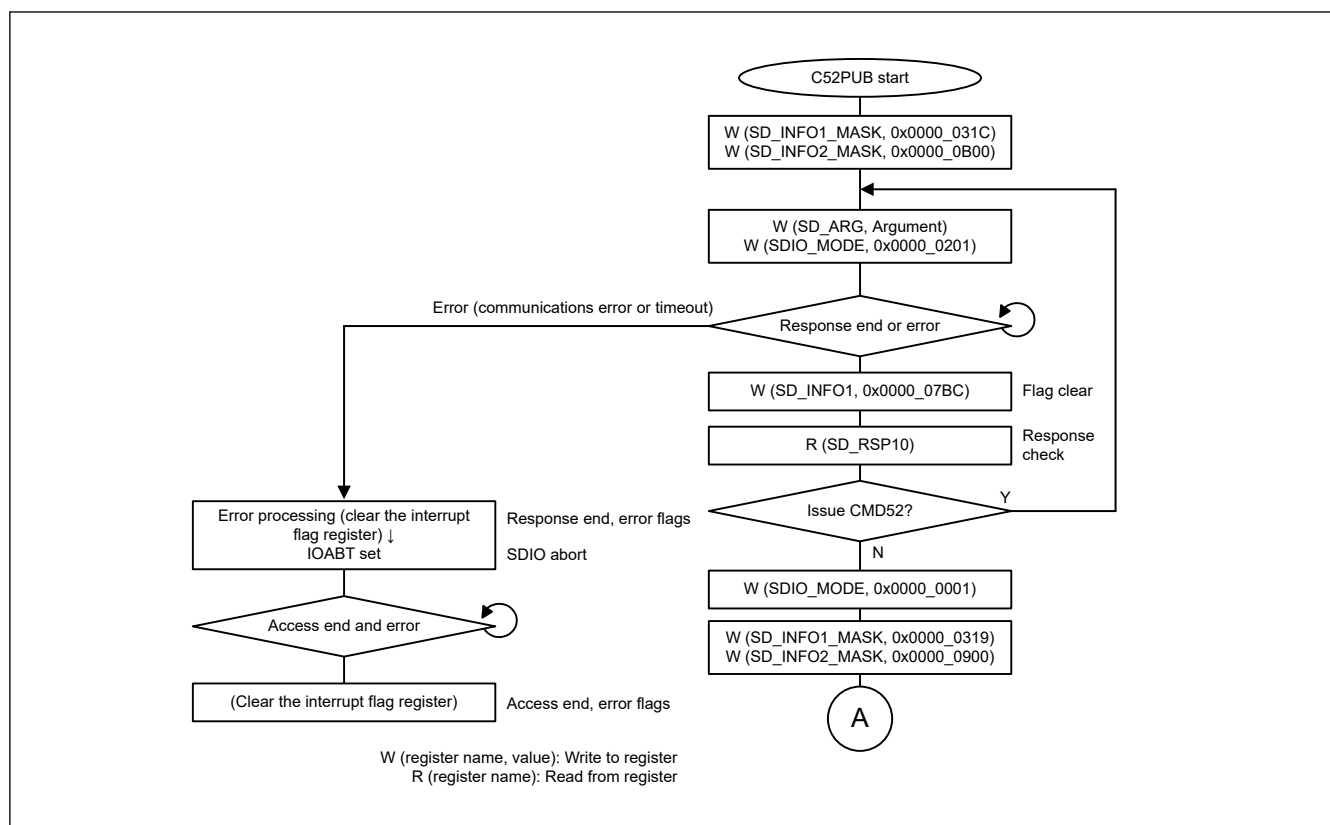


Figure 48.22 Flow when CMD52 (SDIO no abort) is issued during a CMD53 multiple block write

48.3.14 DMA Transfer (SD/MMC)

48.3.14.1 SD_BUF DMA transfer

Figure 48.23 shows an example flow for SD_BUF DMA read when CMD18 multiple block read is issued.

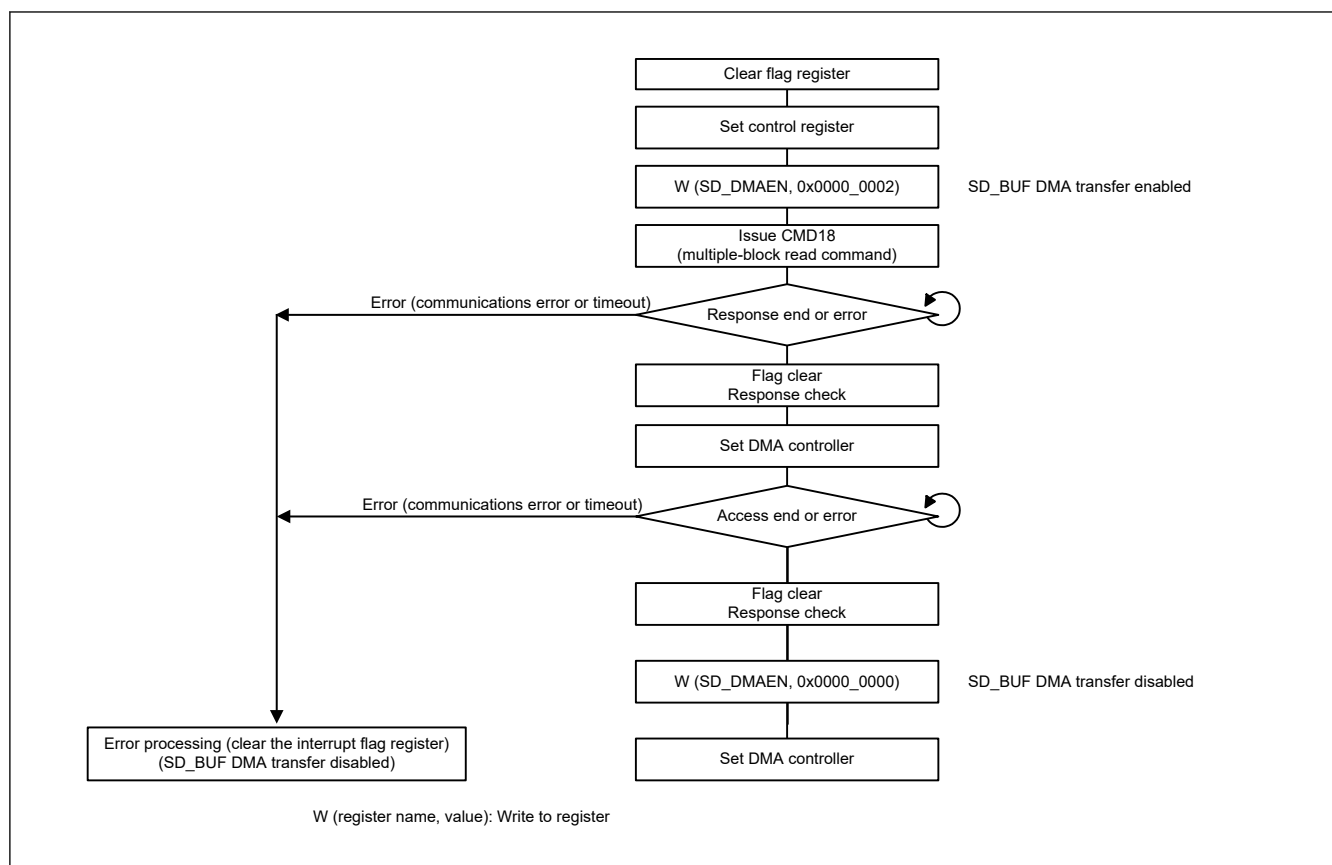


Figure 48.23 Example of SD_BUF_DMA read operation

Figure 48.24 shows an example flow for SD_BUF DMA write when CMD25 multiple block write is issued.

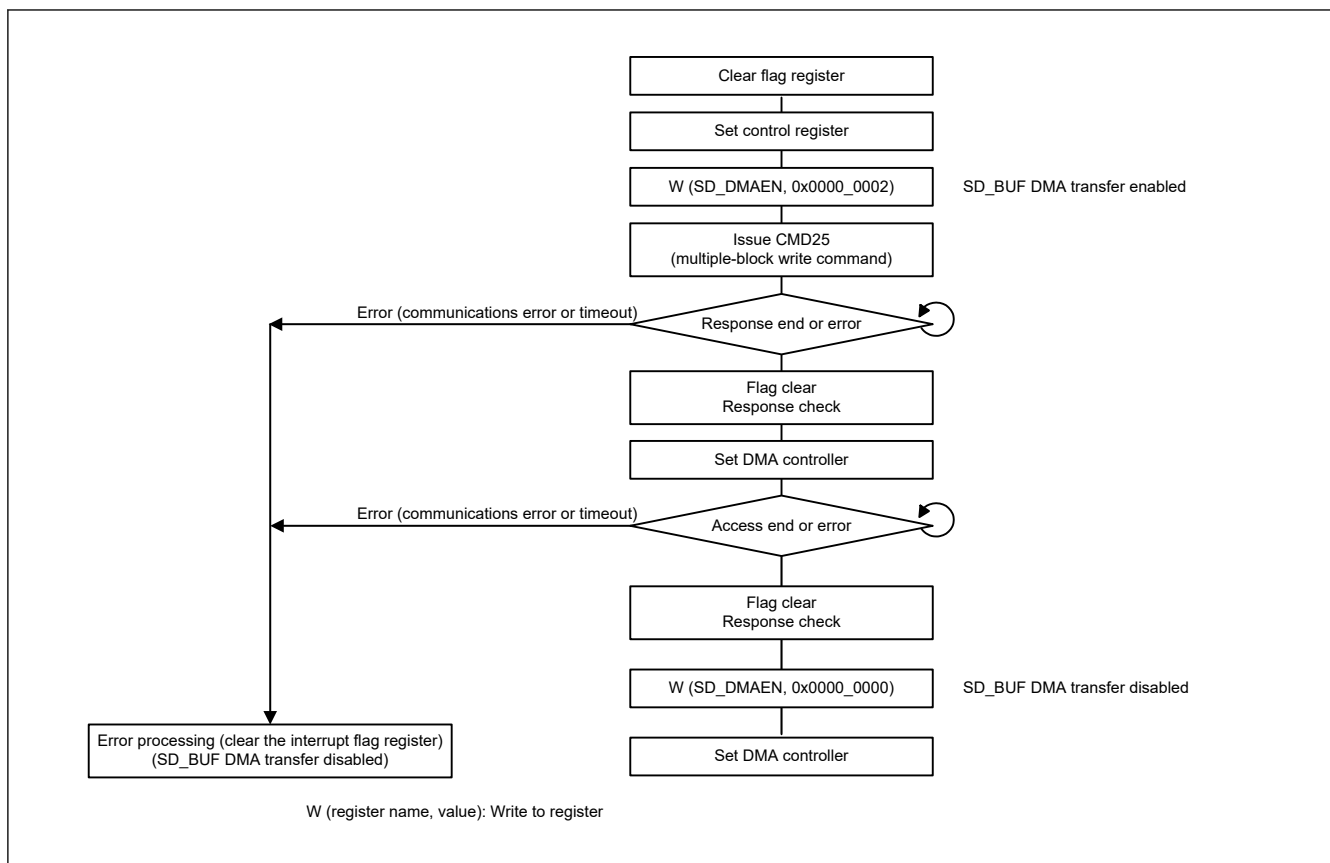


Figure 48.24 Example of SD_BUF_DMA write operation

48.3.15 Example of SD_CMD Register Setting

Table 48.8 and Table 48.9 list the SD_CMD register setting.

Table 48.8 Example SD_CMD register settings for SD (1 of 2)

Type	Command	Example SD_CMD register setting	Remark
CMD	CMD0	0x0000_0000	—
	CMD2	0x0000_0002	—
	CMD3	0x0000_0003	—
	CMD4	0x0000_0004	—
	CMD5	0x0000_0705 or 0x0000_0005	—
	CMD6	0x0000_1C06 or 0x0000_0006	—
	CMD7	0x0000_0007	When the card is placed in the deselected state, the response timeout flag sets because there is no response.
	CMD8	0x0000_0408 or 0x0000_0008	—
	CMD9	0x0000_0009	—
	CMD10	0x0000_000A	—
	CMD11	0x0000_040B or 0x0000_000B	—
	CMD12	0x0000_000C	—
	CMD13	0x0000_000D	—
	CMD15	0x0000_000F	—
	CMD16	0x0000_0010	—
	CMD17	0x0000_0011	—
	CMD18	0x0000_0012	With automatic CMD12
	CMD20	0x0000_0514 or 0x0000_0014	—
	CMD24	0x0000_0018	—
	CMD25	0x0000_0019	With automatic CMD12
	CMD27	0x0000_001B	—
	CMD28	0x0000_001C	—
	CMD29	0x0000_001D	—
	CMD30	0x0000_001E	—
	CMD32	0x0000_0020	—
	CMD33	0x0000_0021	—
	CMD38	0x0000_0026	—
	CMD42	0x0000_002A	—
	CMD52	0x0000_0434 or 0x0000_0034	—
	CMD53	0x0000_1C35	Single read
		0x0000_0C35	Single write
		0x0000_7C35	Multiple read
		0x0000_6C35	Multiple write
		0x0000_0035	The value on the left can be set for both single and multiple operations. However, the CF39 bit in SD_ARG must be set as follows. Read: 0 Write: 1
	CMD55	0x0000_0037	—
	CMD56	0x0000_0038	—

Table 48.8 Example SD_CMD register settings for SD (2 of 2)

Type	Command	Example SD_CMD register setting	Remark
ACMD	ACMD6	0x0000_0046	—
	ACMD13	0x0000_004D	—
	ACMD22	0x0000_0056	—
	ACMD23	0x0000_0057	—
	ACMD41	0x0000_0069	—
	ACMD42	0x0000_006A	—
	ACMD51	0x0000_0073	—

Table 48.9 Example SD_CMD register settings for MMC

Type	Command	Example SD_CMD register setting	Remark
CMD	CMD0	0x0000_0000	—
	CMD1	0x0000_0701	—
	CMD2	0x0000_0002	—
	CMD3	0x0000_0003	—
	CMD4	0x0000_0004	—
	CMD5	0x0000_0505	—
	CMD6	0x0000_0506	(with response busy)
		0x0000_0406	(without response busy)
	CMD7	0x0000_0007	When the card is placed in the deselected state, the response timeout flag sets because there is no response.
	CMD8	0x0000_1C08	—
	CMD9	0x0000_0009	—
	CMD10	0x0000_000A	—
	CMD12	0x0000_000C	—
	CMD13	0x0000_000D	—
	CMD14	0x0000_1C0E	Required setting: SD_IFMODE = 0x0000_0100 (CRC check is invalid)
	CMD15	0x0000_000F	—
	CMD16	0x0000_0010	—
	CMD17	0x0000_0011	—
	CMD18	0x0000_7C12	Pre-defined
	CMD19	0x0000_0C13	Required setting: SD_IFMODE = 0x0000_0100 (CRC check is invalid)
	CMD21	0x0000_1C15	DDR mode is inhibited
	CMD23	0x0000_0017	—
	CMD24	0x0000_0018	—
	CMD25	0x0000_6C19	Pre-defined
	CMD26	0x0000_0C1A	—
	CMD27	0x0000_001B	—
	CMD28	0x0000_001C	—
	CMD29	0x0000_001D	—
	CMD30	0x0000_001E	—
	CMD31	0x0000_1C1F	—
	CMD35	0x0000_0423	—
	CMD36	0x0000_0424	—
	CMD38	0x0000_0026	—
	CMD39	0x0000_0427	—
	CMD40	0x0000_0428	—
	CMD42	0x0000_002A	—
	CMD49	0x0000_0C31	—
	CMD53	0x0000_7C35	—
	CMD54	0x0000_6C36	—
	CMD55	0x0000_0037	—
	CMD56	0x0000_0038	—

48.4 Usage Notes

48.4.1 SD_BUF Illegal Write Access (SD/MMC)

When writing data to SD_BUF0 after the single block write or multi block write command is issued, the data of the size specified in SD_SIZE must be written.

If the data exceeds the size specified in SD_SIZE is written, the ERR4 bit in SD_INFO2 is set to 1. In addition, the data written to SD_BUF0 might not be transmitted and the SD_CLK_CTRLLEN bit in SD_INFO2 is held at the value of 0. If this occurs, clearing the SDRST bit in SOFT_RST to 0 and then restoring its value to 1 clears the SD_CLK_CTRLLEN bit to 1.

However, this does not apply to the single byte or three bytes when the SD_SIZE setting is odd, or to the fraction of bytes when the SD_SIZE setting is even (the 2 bytes that are not in a 4-byte unit), because the portion of dummy data writing is regarded as excess data and ignored.

48.4.2 Block Number Constraint for Multiple Block Read (SD)

When performing a multiple block read of one or two blocks, depending on the timing with which the SD card response register is read, the response value might not be read properly. To prevent this, do one of the following:

1. When receiving one or two blocks of data, use single block reading.
2. Read the response to CMD18 from SD_RSP54.

48.4.2.1 Mechanism of Incorrect Reading

Figure 48.25 shows the processing flows of the SDHI (hardware) operation and software operation when a multiple block read is performed on two blocks. As shown in the incorrect operation in Figure 48.25, when an interrupt is generated on reception of the CMD18 response and the timing with which the SD card response register (SD_RSP10) is read by the interrupt is delayed, the data during the CMD12 response reception or the CMD12 response might be read. This problem does not occur for multiple block reads of three or more blocks, because CMD12 is not issued until the block of data is read. The problem also does not occur for multiple block writes, because the CMD25 response is read before the block of data is sent.

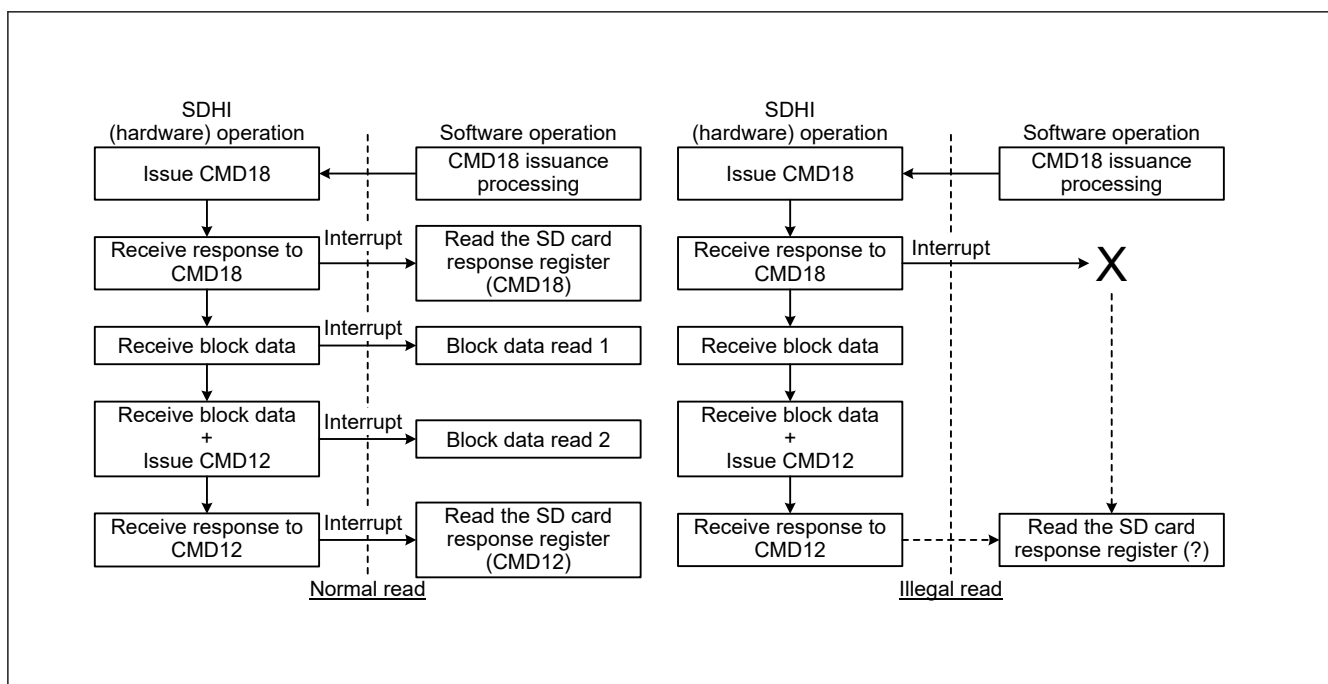


Figure 48.25 Multiple block read operation flow chart (two blocks)

48.4.3 Automatic Control of SD/MMC Clock Output (SD/MMC)

In the SD Card/MMC standard, 74 cycles of SD/MMC clock must be output before initialization of the card. For this reason, use automatic control of SD/MMC clock output after 74 cycles of SD/MMC clock are output. In addition, if

automatic control of SD/MMC clock output was in use, SD/MMC clock output is stopped on completion of the sequence for a communications error or timeout. When state transitions within the SD card/MMC are necessary after completion of the sequence, release automatic control of SD/MMC clock output and restart supply of the SD/MMC clock to the SD card/MMC.

48.4.4 Control of the C52PUB Setting for Multiple Block Write (SD)

If the C52PUB bit in SDIO_MODE is set to 1 during a sequence of multiple block write because of CMD53, CMD52 is not issued until SD_BUF becomes empty. For this reason, set the C52PUB bit after suspending writing to SD_BUF by using one of the following procedures, as appropriate:

(a) When DMA transfer is not in use

1. Before setting the C52PUB bit, suspend writing to SD_BUF by making the setting in SD_INFO2 to disable BWE interrupts.
2. Set the C52PUB bit in SDIO_MODE to 1 (so that CMD52 is issued when SD_BUF becomes empty).
3. After the RSPEND interrupt processing in SD_INFO1 because the issuing of CMD52 is completed, restart writing to SD_BUF by making the setting in SD_INFO2 to enable BWE interrupts.

(b) When DMA transfer is in use

1. Every time DMA transfer of the value set in SD_SIZE $\times$ n blocks (where n = 1, 2,...) proceeds, suspend writing to SD_BUF by DMA transfer before the C52PUB bit is set.
2. Set the C52PUB bit in SDIO_MODE to 1 (so that CMD52 is issued when SD_BUF becomes empty).
3. After the RSPEND interrupt processing in SD_INFO1 because the issuing of CMD52 is completed, restart writing to SD_BUF by DMA transfer.

48.4.5 Notes on SD_CLK_CTRL Register Settings (SD/MMC)

When the SD_CLK_CTRLLEN bit in SD_INFO2 is 0, SD_CLK_CTRL cannot be written to. Before writing to SD_CLK_CTRL, you must check that the SD_CLK_CTRLLEN bit in SD_INFO2 is 1.

48.4.6 Specification Limitations

1. The Suspend/Resume operation of the SDIO is not supported.
2. The SPI bus is not supported. (SD/MMC)
3. The shared bus and 8-bit SD bus of the embedded SDIO are not supported.
4. Stream transfer of MMC is not supported.
5. High Priority Interrupt (HPI) of MMC is not supported.
6. Boot Operation/Alternative Boot Operation of MMC is not supported.
7. Open-ended multiple block transfer of MMC is not supported.

48.4.7 STP Bit Setting During Multiple Block Read (SD/MMC)

During execution of multiple block read with automatic CMD12 execution by setting the SEC bit in SD_STOP to 1, even if the STP bit in SD_STOP is set to 1 to forcibly stop the execution, the command sequence might not stop depending on the timing of setting the STP bit.

To avoid this, when setting the STP bit in SD_STOP to 1 during multiple block transfer, clear the SEC bit in SD_STOP to 0 at the same time. (Even when the SD_CLK_CTRLLEN bit in SD_INFO2 is 0, change the SEC bit from 1 to 0.)

When the command sequence does not stop because the SEC bit was not cleared to 0, the command sequence can be stopped by clearing the SDRST bit in SOFT_RST to 0.

When forcibly terminating the CMD53 multiple block transfer through the IOABT bit in SDIO_MODE, you must leave the SEC bit in SD_STOP as 1.

48.4.8 Register Setting Notes

1. All registers in [section 48.2. Register Descriptions](#) are accessed in 32-bit access-only.
2. When setting registers, set them after the I/O Port Register setting.

48.4.9 Module-stop Function

SDHI operation can be disabled or enabled using Module Stop Control Register C (MSTPCRC). The SDHI module is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).

48.4.10 Software Standby Mode

Before entering Software Standby mode, confirm the SD_INFO2.CBSY bit is 0.